AI Accelerators

#AI

Companies: Nvidia, AMD, Avago, MediaTek, TSMC, ASE, Amkor

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2027 Accelerators Part II – Key Changes

This note addresses key discussions after we released our 2027E accelerator shipments on August 12. In this report, we cover topics including 1) how to justify the compute power gap between chip production and infrastructure readiness in 2027E, 2) rack design trends from heavy and power to small and light, 3) token consumption trends from frontier models to open-source models, and 4) value proposition shifts for ASIC partners. Combining of all these, our preferred picks in the accelerator space are MediaTek and NVDA. MTK benefits from its TPU share gain, and we believe its NVLink-Fusion licensing is critical for it to get new ASIC customers. NVDA benefits from the growth of open-source models, and its DSX platform should expand its unit shipments cap by up to 40% vs what the TDP suggests. In the near term, it also benefits from a slowdown in non-TPU ASICs, while in the long term it can leverage MTK to get more networking business for ASICs. We also rate AMD as a BUY, but it is more of a trading stock to us. AMD is a key winner for agentic AI, but its richest valuation amongst its peers echoes the Street's high expectation, while near-term numbers are supply constrained. Lastly, we believe AVGO is underappreciated. Its non-TPU ASIC gain, mainly OAI, has a good chance of keeping its rapid growth going into 2028E at least, but TPU unit and content loss starting with TPUv8 may keep up the pressure on its P/E multiple.

TDP: peak power vs over-installation

The aggregate compute power by 2027E-produced accelerators will reach ~55GW, significantly higher than the infrastructure accommodability of the <mid-40GWs if everything goes smoothly. One can offer several reasons to justify the gap, including behind-meter power supply and time differences from chip production to DC installation. We think another key factor to explain the gap is power oversubscription (or chip over-installation). NVDA's DSX standardizes it and allows DC operators to install up to 40% more GPUs than suggested by TDP. CSPs such as MSFT and Google and chipmakers such as AMD also have their own dynamic power management systems to maximize chip installation units and power efficiency.

Smaller racks, bigger pods

Contrary to the conventional wisdom that all chips and racks are growing bigger and more powerful, we note chipmakers are limiting their rack solutions from getting bigger and consuming more power. Power supply, thermal dissipation and weight are the three key factors that force chipmakers and customers to do so. Networking will take a more important role to construct a cluster with more accelerators. We have also observed similar trends for Chinese accelerators at WAIC 2026, given the limitation on them on producing more powerful chips.

Token consumption: rapid growth of open-source models

Token consumption by open-source models is growing at a much faster pace than those closed frontier models. Agentic swarm, TTC and COT, synthetic data generation and distilling, and CI/CD will all boost token output dramatically. Token cost from frontier models is much more expensive and/or more restrictive, which pushes users to enable more open-source model usage. Literally all open-source models are run on NVDA GPUs.

ASICs: front-end design vs back-end and system integration

OAI's Jalapeno experience suggests compute front-end design, though the core of any ASICs, is getting easier thanks to AI. On the other hand, I/O, advance packaging, networking, system design and supply chain management are what those ASIC customers, especially AI model labs, are lacking. Traditional compute front-end design could play a less critical role for ASIC designers in the winning of projects in the future.

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Infrastructure availability: are we producing too many GPUs/ASICs in 2027?

The aggregate compute power for 2027 ASICs and GPUs, based on our production estimates, will reach 53GW. If we include those not in our estimates such as OAI's Jalapeno and MSFT's Maia 200, the number could surge to 55GW+. On the contrary, total infrastructure readiness in the US for AI data centers is estimated at about 30GW if all planned projects go smoothly, and US is considered to be at least 70%+ of the global market excluding China. This leads to a general question from investors about whether too many accelerators will be produced in 2027E. We believe the answer is no. In addition to the time lag (that is, chips produced today are for installation in two to three quarters down the line) and behind-meter power supply, the critical reason is that real chip installation is usually higher than the TDP suggests. In the next paragraphs, we will use NVDA's DSX as an example to explain why the "over installation" approach works in the real world.

FIGURE 1: 2027E GPU/ASICS PRODUCTION NUMBERS AND IMPLIED COMPUTE POWER

	Chip Vol* (m)	TDP (W)	PUE^	Compute / Rack	Chip Vol per GW (k)	Implied Power (GW)
NVDA	**9.5**					**33.4**
B200/B300	1.0	1,300	1.1	70%	486	2.1
Rubin	6.2	2,300	1.1	70%	275	22.6
Rubin Ultra	2.3	2,400	1.1	70%	263	8.7
AMD	**1.6**					**4.9**
MI450X/455X**	1.1	2,500	1.2	70%	232	4.7
MI550X	0.0	4,500	1.2	70%	129	0.2
Google	**8.2**					**12.0**
TPUv8i (Sunfish)	4.0	1,100	1.1	70%	574	7.0
TPUv8t (Zebrafish)	4.0	700	1.1	70%	903	4.4
TPUv8i+ (Whalefish)	0.2	2,000	1.1	70%	316	0.6
AWS	**2.0**					**2.2**
Trainium 3	2.0	650	1.2	70%	891	2.2
TOTAL	**21.3**					**52.6**

Source: Aletheia TAG *estimated production and could differ from actual shipments ^1.1 for NVDA and TPU; 1.2 for others **Assuming 2x MI450X (Meta version) equals to 1x MI455X (standard) ***theoretical volume defined by TDP, actual volume per 1GW could differ significantly subject to DC design

NVDA's DSX platform: MaxP, MaxQ, and MaxLPS

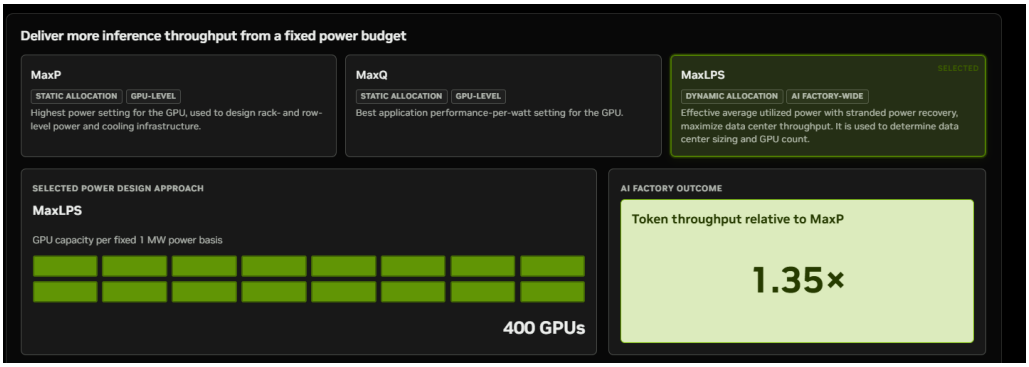
NVDA's DSX is its reference design for AI factories starting from the GB200 NVL72. It covers from platform management software to hardware rack installation layout. The key idea is that not all GPUs will run at their peak power all the time, with NVDA identifying three different power settings: MaxP (Max Performance), MaxQ (Max Efficiency), and MaxLPS (Max Land Power Shell). The former two are static power settings at the GPU level while the last one refers to dynamic power allocation at the AI factory level (which will also lead to the lowest power consumption at GPU/rack level).

Without going into detail, NVDA claims that MaxQ and MaxLPS status can generate the best token cost for most applications. Even training can be done most efficiently under MaxQ rather than MaxP. Given GPUs are running at much lower power compared with MaxP, which is running at its TDP (Thermal Designed Power), every 1GW of infrastructure will be able to house more GPUs if the data center is designed under MaxQ or MaxLPS. Every 1GW can accommodate 275k Rubin GPUs under Vera-Rubin NVL72 architecture at a PUE of 1.1 (38.2k racks). The number of GPUs per 1GW will increase to 330k/400k under MaxQ and MaxLPS design, representing 20%/46% more GPUs per 1GW. NVDA disclosed details here.

NVDA will continue this DSX offering and we already have some initial color for the Rubin Ultra generation. We note that rack power for the Rubin Ultra NVL72 is 235kW under a MaxP configuration, which is just a small increase from 225kW for Vera Rubin NVL72. Under MaxQ and MaxLPS, rack power will go down to ~170kW and ~140kW. For the GB300 NVL72, MaxP rack power is 150kW and MaxLPS is about 100kW. This suggests the Rubin Ultra NVL72 could

consume lower power vs the GB300 if the former runs at MaxLPS status vs MaxP for the latter. Also, DC operators can use software control to freely switch among these modes, which provides DC operators the flexibility to run their AI factories at the most efficient mode for different demands. The bottom line is – DC operators could install more GPUs vs the theoretical number of GPUs defined by TDP, and such "over installation" can be as much as 45% in the case of NVDA's Vera Rubin NVL72.

FIGURE 2: NVDA'S MAXLPS ALLOWS 35% MORE GPU CAPACITY VS WHAT TDP SUGGESTED



Source: Nvidia *for Vera Rubin NVL72 inference scenario; per fixed 1MW

The "power oversubscription" concept is not new

While NVDA's DSX reference solution quantifies the number of GPUs that can be over-installed in a given power supply, the concept of power oversubscription is nothing new. It is already common for general server DCs to install CPUs up to 106% of what the TDP suggests because all CPUs rarely run at their peak speed together. DC operators also have their own proprietary approach to maximizing chip over-installation. By doing so, power infrastructure capex as a percentage of total spending can be effectively reduced. For instance, MSFT has its own workload VM prediction system, which allows it to be over-install 12%+ CPUs vs what the theoretical TDP suggests, 2x the conventional 6%, in its Azure cloud. This approach effectively reduced up to $75m capex for power infrastructure for 128MW DC. Going into the AI era, MSFT has also announced its nvPAX dynamic power management system to maximize chip installation and power efficiency. Other than MSFT, Meta and Google also have their own dynamic workload prioritization system for the same purpose. At the chipmaker level, we have also learnt that AMD's server CPU and AI GPU both support its power capping feature that allows its CSP customers to customize their own chip workload management systems.

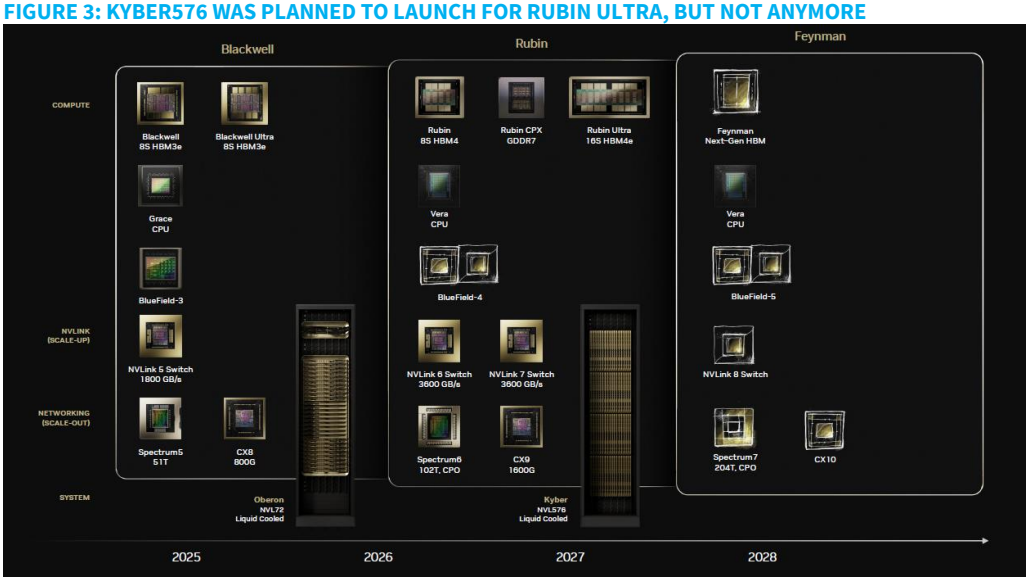
Smaller racks, bigger pods

Two recent developments caught our eye about AI server rack design. The first is NVDA's Kyber delay, and the second is the absence of AMD's Titan rack.

NVDA: only modest TDP increase from Vera-Rubin to Rubin Ultra rack

NVDA unveiled its Kyber576 rack for the first time at GTC 2025. The design uses "blade" server boards to fill in up to 576x GPU cores, or 144x 4-tile Rubin-Ultra GPUs. Assuming NVDA can keep GPU compute at 70% of rack power, this means the TDP per rack will be as high as 1MW assuming the TDP for the 4-tile Rubin Ultra is 5,000W. This is almost 4.5x the TDP for Vera-Rubin NVL72 of 225kW.

FIGURE 3: KYBER576 WAS PLANNED TO LAUNCH FOR RUBIN ULTRA, BUT NOT ANYMORE



Source: Nvidia

The real world is another story. The Kyber rack is delayed, and it is already well known it will not be in time for Rubin Ultra. The Rubin Ultra's rack is most likely to continue the current Oberon structure and use CPO/NPO for inter-rack scale-up to construct a 576-GPU pod instead of stuffing everything into a single rack under the Kyber structure. In the meantime, Rubin Ultra is also most likely to start with a 2-tile solution with either HBM4 8H or HBM4E 8H. Under a MaxP configuration for the HBM4E 8H version, we estimate the Rubin Ultra rack TDP is just 235kW, only modestly higher than the 225kW for the Vera-Rubin rack.

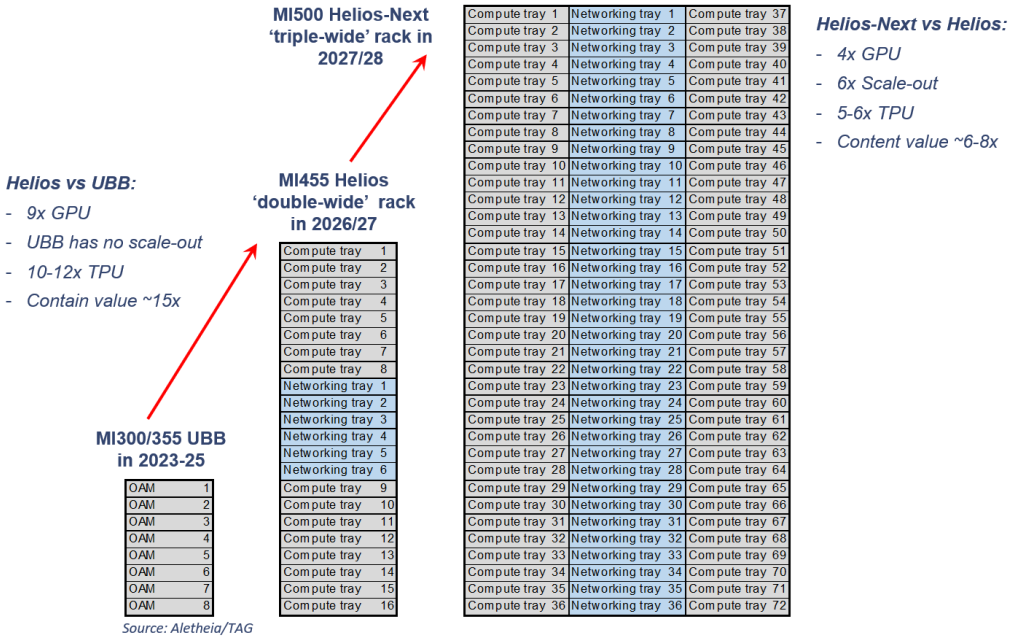
AMD: the absence of Titan rack

Our prior research has suggested that AMD planned to introduce the Titan rack for the MI550X AI GPU, which is expected to go into HVM before end-2027. The Titan rack is designed as 3x the width of the conventional server rack. Its predecessor, Helio rack, which houses MI455X GPUs, is only 2x the width vs conventional. The Titan rack will consist of 72 compute trays and 36 networking trays compared with 18x/6x for Helio racks. Each compute tray in the Titan rack will house 4x MI550X GPUs, suggesting total of 288x MI550X GPUs in a rack. We estimate the TDP per MI550X GPU is as high as 5000W. Should GPUs also account for 70% of rack power (similar to NVDA), the TDP for the Titan rack will reach 2MW+.

At its latest AI event in July 2026, however, AMD didn't mention the Titan rack anymore. It only reiterated the Helio rack at the event. We understand that there will be multiple rack designs going into the MI550X generation, including the Helio rack and the conventional width versions. The MI550X will also have at least two SKUs: the full-sized one (8x compute die, 4x base-die, 4x I/O dies and 16x HBMs) will be available in 1H28E, and the half-size version (4x compute dies, 2x base-dies, 2x I/O dies and 8x HBMs) will go into HVM first in 2H27E. Rack

solutions for the half-sized one will still be based on the Helio rack and its TDP should be similar to the ~250kW of the Helio rack for the MI455x. On the other hand, the full-sized MI550X rack solution will go back to the conventional width, and each rack will only carry 16x or 32x GPUs, making the TDP at 100kW+ or 200kW+. AMD will also have a 512x GPU pod, which uses optical inter-rack scale to connect 512x full-sized MI550X, that is, 16x or 32x racks, together. This approach is very similar to NVDA's NVL576, which uses CPO or NPO to connect 8x Rubin-Ultra NVL72 racks together.

FIGURE 4: THE ORIGINAL PLAN FOR AMD'S MI500X RACK "TITAN" – NO LONGER THE CASE



Source: AMD, Aletheia/TAG

Infrastructure accommodability and heat dissipation

We believe the infrastructure accommodability and heat dissipation issue are the two main reasons why both NVDA and AMD have changed the rack design for their next-generation GPUs. To house bigger racks, AI data center infrastructure not only has to provide higher power density but also needs to be firm enough to carry these heavy racks. Power density has been widely addressed, and will require 800V high voltage power supply, which the industry is still developing and ramping. Thermal is another issue why the entire cooling system will be massively reinvented to accommodate these high power density racks. In fact, we believe the thermal issue is one of the reasons why the Kyber rack is delayed (together with many other reasons such as midplane redesign).

Rack weight is less addressed but is as crucial. For example, the current GB300 rack NVL72 already weighs ~1.6tons and the floor load density needs to be up to 2.2-2.5tons to carry such a rack. If the Kyber rack really materializes, it could be as heavy as 6-7 tons as it carries 4x more compute dies, a midplane for networking and heavier power delivery. AMD's Helio rack is double the width of conventional racks and weighs 3 tons+. That means a similar weight per area space to the GB300 NVL72. Titan racks will carry 4x GPUs than Helio racks, though they are only 1.5x as wide as the Helio (3x conventional). If it is also 4x the weight, it will become ~12tons for 3x the width of a standard rack, or four tons per standard rack width, which is more than 2x than that of the Helio rack. All of these will require a significant design change in AI data center infrastructure as existing sites are unlikely to be able to house these monsters. To be more installation-friendly, we believe most clients are requesting lighter and smaller racks.

Google is the forerunner of this trend; Chinese are doing similar

While smaller and lighter racks provide better accommodability, less power and fewer thermal issues, it is at the expense of fewer GPUs or ASICs in each rack. This may sacrifice the efficiency for those workloads such as training that need multi-GPUs. This should explain why NVDA and AMD are working on inter-rack scale-up (or some call it scale-across) to form a bigger pod without physically stuffing these GPUs into one rack. In fact, Google is the forerunner of this approach. For instance, its TPUv7 (Ironwood) only carries 64 TPUs in a rack, but it uses OCI to connect up to 144x TPU racks (a total of 9.216 TPUs) into one superpod. When it goes into TPUv8t, TPUs per rack stay at 64 but the total racks connected increases to 150 and the superpod consists of 9,600 TPUs. TPU8i even carries a small amount of TPUs in one rack (32x) but can also connect up to 36 racks in one superpod. In our visit to WACC (World AI Congress) 2026 in Shanghai in July, we also learned that all Chinese accelerator suppliers are pushing more than 10k accelerators superpod to make up for the disadvantage of lower compute power per chip given their limited access to advanced semiconductor processes. We believe international chipmakers are moving toward a similar trend at least in their next-generation products.

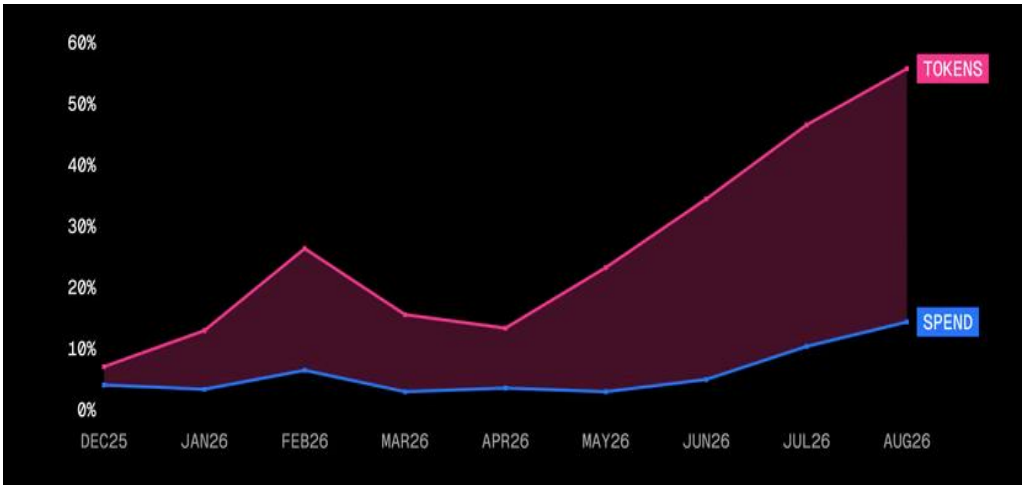
FIGURE 5: RACK AND POD DESIGN FOR SELECTED UPCOMING GPUS AND ASICS

	Rubin Ultra (dual-die)	MI500X (quad-die)*	TPUv8t	Open AI
Scalability	576	512	1,152	2,048
Scale-up networking	QM5	TH/T100/S1G300	ICI	TH6
System Compute	29EFLOPs+	51/102EFLOPs	12EFLOPs	27EFLOPs
Chips per rack	72	16/32**	12EFLOPs	6
Racks per pod	8	32/16	36	16
Rack TDP	235kW	100/200kW+	70-80kW	~125kW
Rack HVM	2H27	1H28	1H27	2027

Source: Company, Aletheia Capital/TAG *quad-die refers to 4x base-die; there is another dual-die version which will come earlier and go to Helio rack **detailed configuration is not finalized yet

Token consumption

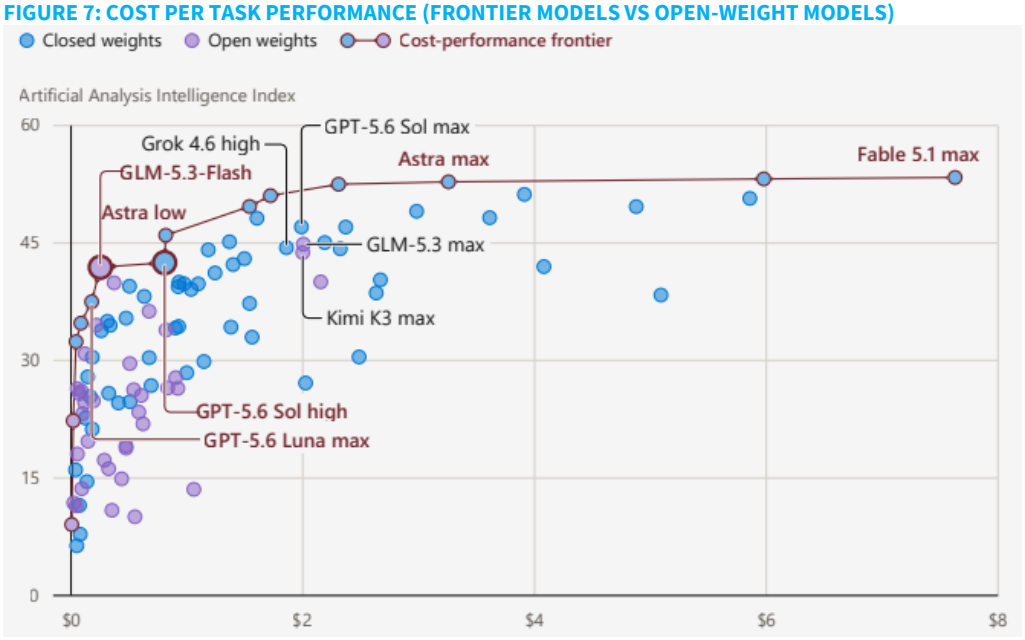
FIGURE 6: OPEN-WEIGHT SHARE OF TOKENS AND SPEND ON AI GATEWAYS



Source: Vercel; AI Gateway

There is a fundamental misconception about how the market sees or analyzes token demand. The market treats all token demand as a uniform curve driven by human-prompted chatbots. While this might be true for the frontier models like GPT-4 and Claude, which are constrained to low velocity, open-source models (open-weight is a more accurate term) are absorbing high velocity, automated machine-to-machine workloads. According to OpenRouter, a major routing platform, open-source models accounted for 75% of all token usage on the OpenRouter platform in Q3 2026, which is up from 30% in late 2025, whereas it reached 62% for Vercel's AI Gateway platform in August 2026. Note, these figures are not global market share as these numbers do not contain direct usage of closed frontier models. Nevertheless, this shows a broader trend in the direction of where the markets are moving at the onset of Agentic AI development. Open-source models are more economical than closed frontier models because they eliminate proprietary markup fees, allow self-hosting, and match 90% of routine enterprise capabilities at a fraction of the cost.

FIGURE 7: COST PER TASK PERFORMANCE (FRONTIER MODELS VS OPEN-WEIGHT MODELS)



Source: Microsoft AI Economy Institute; Artificial Analysis

We discuss some important topics that explain the cause of this structural divergence in token Consumption between open-source and frontier models.

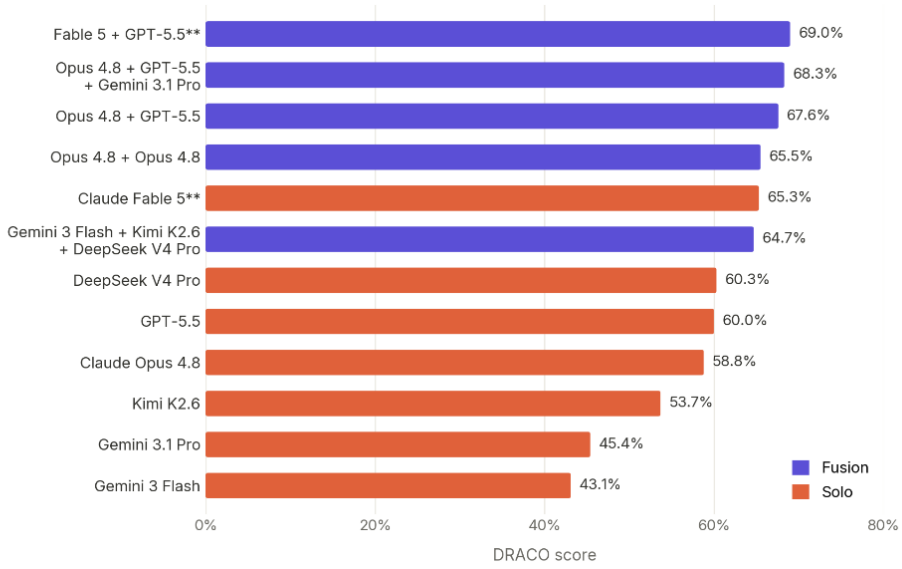
The multiplier effect caused by "Agentic Swarm"

An agentic swarm is a decentralized multi-agent AI system where autonomous, specialized AI agents collaborate, divide tasks and adapt in parallel to achieve a shared goal. Instead of relying on a single large language model (LLM) or a strict hub-and-spoke pipeline, an agentic swarm acts as a force multiplier by executing sub-tasks in parallel. When moving from a single AI assistant to a distributed swarm of dozens or hundreds of parallel agents, frontier API models become logistically, financially and architecturally restrictive. The token consumption in an agentic swarm AI system increases multifold. Frontier API charge explicitly per 1mn input and output tokens. If an enterprise deploys an agentic framework using a frontier API model, a single complex task might require hundreds of sub-queries as the agent debates with itself, retrieves data and corrects its own errors (a loop). This compounds the API costs exponentially, making agentic architectures on frontier models financially unviable at scale.

In case of self-hosted open-source models (only open-source models can be locally hosted by the user, frontier models are locked in this sense), the marginal cost collapses only to the raw cost of power and hardware depreciation once the server is provisioned and the enterprises can use open-source models as the default engines for agentic loops. On top of that, recent data suggest that agentic token usage is vastly larger than single-shot usage. Automated agents consume exponentially more tokens continuously in the background without human initiation.

Using multi-agent AI systems enables enterprises to combine frontier models with open-source models to get better performance and optimize the cost compared to a standalone frontier model. Figure 8 shows benchmarking of OpenRouter's fusion platform with different models and combinations. We can see Gemini 3 (frontier model) + Kimi k2.6 + DeepSeek V4 Pro (open-source models) scores 64.7%, much higher than Gemini 3 Flash at 43.1%.

FIGURE 8: DRACO BENCHMARKING OF OPENROUTER FUSION TOOL



Source: Open Router

The industry's recognition of this API cost ceiling was definitively validated in 3Q26 with Meta's dual-release strategy for its Muse agentic ecosystem. While the company deployed its proprietary API, Muse Spark, charging up to $4.25 per 1mn output tokens for standard tier usage, they simultaneously released Muse Glimmer, an open-weight 30-billion parameter

model under an Apache 2.0 license. Glimmer was explicitly engineered to run entirely offline on a single 24GB consumer GPU for 'local agentic and coding workloads'. This move effectively conceded that for developers to execute continuous, high-volume agentic loops—where an AI autonomously drives a browser, navigates systems, and executes sub-tasks—the marginal cost must be driven to zero through self-hosted open-weight infrastructure rather than tollgated APIs.

Test-Time Compute (TTC)/Hidden Chain-of-thought (CoT)

The architectural shift towards reasoning models has fundamentally broken traditional token consumption forecasting. Historically, output token volume was roughly proportional to the length of the final answer. The introduction of Test-Time Compute (TTC) models-pioneered openly by architectures like DeepSeek-R1 broke this ratio. TTC models are trained using Reinforced Learning (RL) to think before they answer. During inference, these models generate a massive, hidden sequence of reasoning tokens to plan, correct and evaluate strategies before outputting the final use-facing response. A prompt that previously required 200 output tokens on a standard dense model might now generate 10,000 to 30,000 hidden reasoning tokens before delivering 200-token answers.

Here again the economic cost of inference comes into play, frontier models factor in a premium in API rates to cover this extended compute time, whereas users can aggressively fine-tune open-source models to run on self-hosted hardware and exponentially reduce the costs of inference.

Synthetic Data Generation and Model Distillation

Synthetic Data Generation involves using a highly capable “teacher” model to generate millions of high-quality examples, which are then used to train or fine-tune a smaller, cheaper “student” model (Distillation). The terms of service of major frontier models explicitly prohibit using their outputs to train competing models. This forces developers, startups and enterprises who want to build their own proprietary small language models (SLMs) to use permissive open-source models for distillation. Other than restrictions, generating a synthetic dataset requires running massive, parallelized batches of prompts. According to OpenAI's own documentation, producing high-quality datasets requires multi-stage filtering, document chunking, and iterative evolution, meaning a single raw input can require dozens of LLM passes to produce one "golden" synthetic row. Because this requires hundreds of billions of tokens, the variable cost of an API makes SDG economically lethal, hence, the entire SDG pipeline has migrated to self-hosted, open-weight clusters.

Continuous Integration and Continuous Deployment (CI/CD)

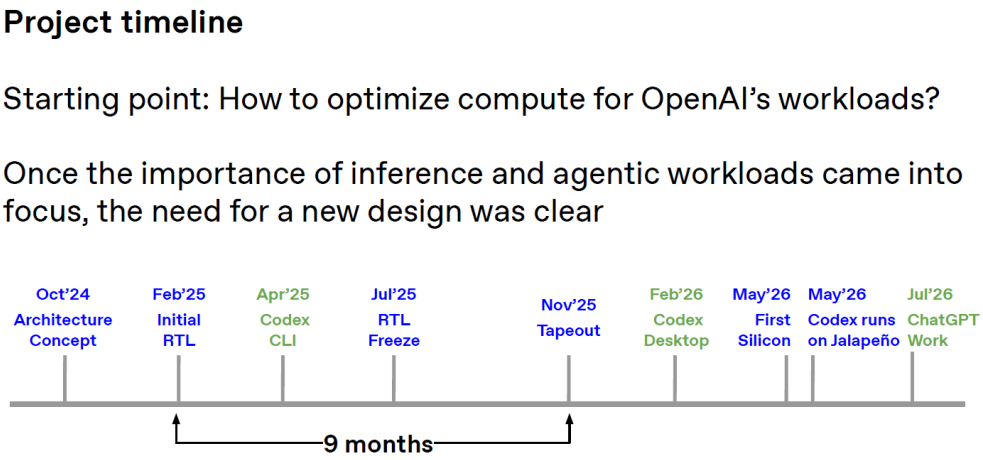
In software engineering, open-source models are now deeply embedded directly into Continuous Integration and Continuous Deployment (CI/CD) pipelines. Instead of a developer manually asking an AI to review code, an open-weight model is triggered automatically on every single keystroke, commits, and pulls requests across an entire enterprise engineering organization. A typical organization may produce thousands of code changes each week. Running automated analysis pipelines that ingest full project context and generate outputs such as static analysis results, security findings, or unit tests for every change can drive computational workloads into the tens of billions of tokens per month. Sustaining this kind of continuous, system-driven volume is only practical on locally hosted, open-weight models where incremental token costs are effectively zero.

ASIC design trends – less front-end, more back-end and integration

OAI showcased how AI speed up the chip design cycle

At Hot Chips 2026, OAI provided more details about its first ASIC, "Jalapeno", for which AVGO is its design partner. OAI suggested it took only nine months from the initial RTL (functional design confirmation) to tapeout, which traditionally needs 12 to 18 months. After tapeout, it usually needs 9-12 months to enter HVM. This means OAI's Jalapeno will be able to enter HVM in 4Q26 (tapeout in November 25), effectively shortening the chip development time from the common two to three years to below two. For instance, we believe Google will finalize its design partner for its TPUv10 in the coming one to two months, but such chips will only enter HBM in 1H29 if everything goes smoothly. Shorter development cycles not only reduce costs but also ensure chips better align with real-time demands, saving developers from trying to guess requirements for the longer term.

FIGURE 9: DESIGN TIMELINE FOR OAI'S JALAPENO



Source: Open AI

Value shift to system integration and back-end design

Another takeaway from OAI's Jalapeno is that the entry barriers for front-end chip design is significantly reduced. We believe OAI only has a team of hundreds, if that, engineers for its chip design, but it can complete the front-end design in such a short period of time. OAI suggested AI helps make the front-end design, especially for the compute tile, much easier and quicker. We believe AVGO, its design partner, also helps a lot here by leveraging its TPU experience. On the other hand, we don't see OAI as being capable of handling I/O tiles (short of SerDes technology), back-end packaging and testing, chiplet system integration, rack architecture, supply management, and so on, given its small-scale team. All these works require its design partner to handle, so AVGO plays an important role here.

If all these CSPs and AI labs can leverage AI tools to lower the entry barriers for front-end design, we believe the value proposition for ASIC design partners should move more to the back-end and system level, rather than the chip front-end design. At the chip design level, I/O and connectivity could still be something that ASIC design partners can provide value for because this is also a capability that many CSPs and AI labs don't have. This is exactly what AWS has been doing for years for its Trainium chips and what Google has started to try in its TPUv8t (Zebrafish), or the so-called COT model (custom own tool).

Figure 10 shows how Google and its design partners are sharing the workload from TPUv8 to TPUv10. Clearly, Google is taking more workloads generation-to-generation, but we note I/O and 2.5D packaging are the two areas where Google still relies on its design partners. AI labs

such as OAI or Anthropic which don't have the experience and physical resources to execute non-design work will rely on their design partners more. What is equally, if not more important, though not highlighted in this table, is system design, from the rack level to the cluster level, which also requires networking expertise.

FIGURE 10: TPU DESIGN PARTNERS AND WORKLOAD ALLOCATION, ACTUAL AND POTENTIALS

Product code	TPUv8t Zebrafish	TPUv8i Subfish	TPUv8.5i Whalefish	TPUv9 Humufish/Triggerfish	TPUv10-M Merope	TPUv10-I Icefish
Compute die	GOOGL	AVGO	AVGO	GOOGL	GOOGL/AMD/AVGO	GOOGL
Sub-compute/Base die	N.A.	N.A.	N.A.	MTK	AVGO	GOOGL
I/O	MTK	AVGO	AVGO	MTK	AVGO	MTK
SoIC 3D Packaging	N.A.	N.A.	N.A.	N.A.	AVGO/AMD	GOOGL/GUC
2.5D Packaging	CoWoS	CoWoS	CoWoS	EMIB-T	EMIB-T	EMIB-T
HBM Procurement	GOOGL	AVGO	AVGO	GOOGL	GOOGL	GOOGL
HVM	4Q26	4Q26	4Q27	1H28	1H29	2029

Source: Aletheia Capital/TAG

Here we try to identify the capabilities of key ASIC designers. We include AMD in the list because it is also bidding for TPUs now. There is no doubt AVGO remains the only all-round design partner for chips itself and networking. It has not delivered any rack-level solutions for its customers, but we suspect OAI's Jalapeno rack has significant AVGO knowhow inside. MTK has got no chance to prove its compute design capability or its networking solutions. By licensing NVLink Fusion, MTK now can provide as many comprehensive solutions as AVGO except compute front-end design – which we argue will see less value in the future. AMD is also lacking I/O and networking by itself but working together with MRVL can make it up. Lastly, we believe Alchip and GUC are still more of suppliers for back-end packaging as well as logistic handlers (such as wafer order placement handling, and so on).

FIGURE 11: KEY ASIC DESIGNERS' CAPABILITIES

	AVGO	MTK	AMD	MRVL	Alchip	GUC
Compute design	v	?	v	?	x	x
I/O	v	v	?	v	o	o
Advanced packaging	v	v	v	v	v	v
Networking	v	o	x	v	x	x
Rack-level design	?	x	v	?	x	x
AIDC system design	x	x	x	x	x	x

Source: Aletheia Capital/TAG v: capable, ? not proven, o gathered from partners, x not capable

MediaTek

↔ Buy

#Semiconductor #5G #Smartphone #AI #TPU #ASIC

Price: **TWD 4,920** Target Price: **TWD 6,500** Upside: **32.1%** Total return: **37.4%** Risk: High

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MediaTek

Bloomberg	2454 TT
RIC	2454.TW
Market cap (USDbn)	246
Average daily T/O (USD|bn)	1.4

Beyond TPUv8

We reiterate a BUY rating for MediaTek (MTK) and raise our PE-based TP to TWD6,500 from TWD5,300, based on 25x average 2027/28E P/E. We are more confident about MTK's potential for TPUv8/9 growth given a surge in both supply and demand. This should continue to drive its strong earnings growth into 2029, and with the likely TPuv10, the momentum can further extend into the 2030s and onwards. We also believe its NVLink Fusion licensing is helpful for winning more ASIC businesses –it is a pure optional value to our forecast and investment thesis. MTK is our preferred AI compute pick.

TPU momentum could continue into 2030E onwards

We are revising up our TPUv8t shipments to 4.2m/4m in 2027/28E driven by increasing supply from TSMC. Its 2027E ASIC revenue will exceed $20b in this case, meaningfully above its $12-$16b guidance. For TPUv9, our checks suggest EMIT-T process yield has improved to 50% recently, on pace for its 80%+ target before HVM. EMIB-T substrate yield is still suppressed but there is increasing supply from new vendors such as AT&S. On the demand side, MTK will expand its SAM to external customers (Anthropic) from Google internal only, starting from its TPUv9, and the momentum will continue into 2029/30E. Its total TPU compute power will reach 16GW+ in 2028E accordingly, up from <5GW in 2027E and exceeding AVGO's 10GW. Lastly, we believe MTK has the best chance to win its TPUv10 project (Icefish) and with better content vs its TPUv9, given its competitive cost structure and improving Serdes technology. We estimate MTK's TPU revenue will see sequential YoY growth for five years starting 2026E.

Impact from NVLink Fusion Licensing

Licensing NVLink Fusion is an important strategic move for MTK. With NVLink Fusion, MTK not only has a full stack of networking solutions but can enable its customers to easily connect their ASICs with NVDA's GPU, providing better scaling flexibility for its customers. Real earnings contributions may only come in 2029 onwards given the design cycle, but we believe lots of client engagement activities are happening now. Literally all frontier model labs and CSPs are customers. We should reasonably expect some new project wins in the next 12 months or so.

Near-term Drivers: TPUv8t shipments and new customer wins

For the near-term, we expect MTK's TPUv8t shipments to kick off from 4Q26E with $2.5b revenue this year vs the $2b+ guidance. Going into 2027E, we also believe MTK has secured Bytedance as its second ASIC customer – though this is more a capacity support angle just like QCOM, we estimate it provides a $1b+ revenue opportunity in 2027E.

KEY FINANCIAL AND VALUATION MATRIX

Dec 31 (TWD bn)	2022A	2023A	2024A	2025A	2026E	2027E	2028E
Revenues	549	433	531	596	680	1,360	2,096
Op. Profit	127	72	102	103	112	316	567
Net Profit	119	77	107	106	117	296	517
EPS ($)	74.19	48.33	66.78	66.11	72.92	185.78	324.22
Consensus EPS ($)	-	-	-	-	68.62	143.25	275.70
PER (X)	66.3	101.8	73.7	74.4	67.5	26.5	15.2
PBR (X)	17.6	21.0	19.4	19.2	17.6	12.1	8.5
Dividend Yield (%)*	1.5	1.5	1.1	1.1	1.1	1.2	3.0
ROE (%)	27.0	18.8	27.3	25.9	27.2	54.2	65.6

Source: MediaTek, FactSet, Aletheia/TAG

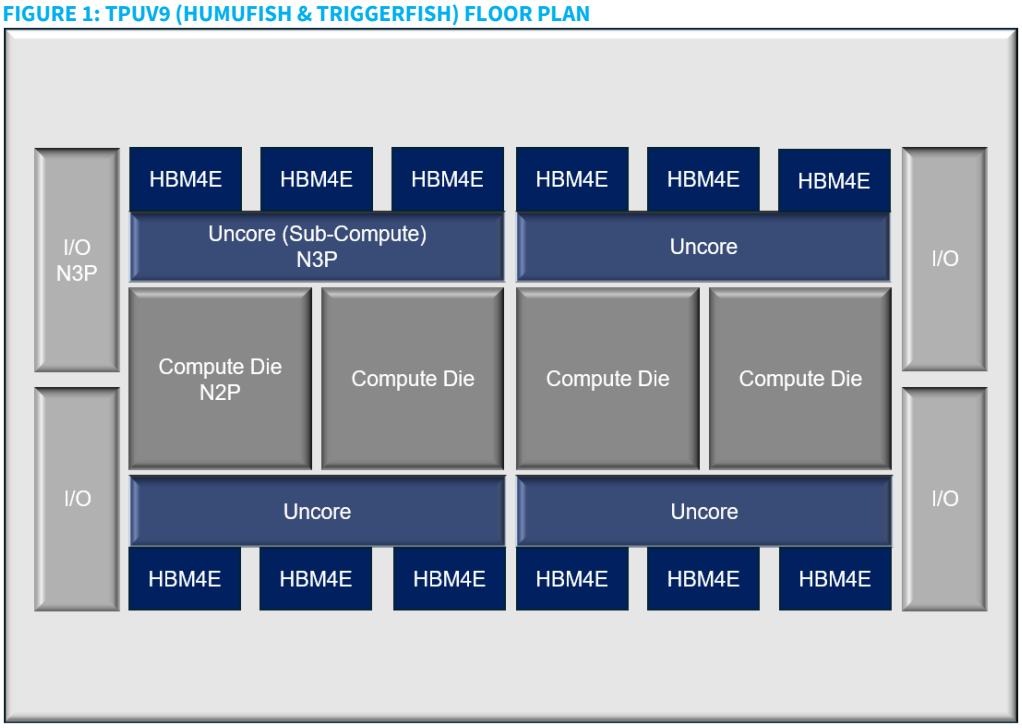
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A closer look at TPU status

A snapshot of TPUv9

Let's start with TPUv9 before we go into detail about TPUv10. There will be two versions of MediaTek's TPUv9: Humufish and Triggerfish. Humufish is for Google's internal use only, while Triggerfish is for Anthropic, which we believe is optimized for inferencing by allocating more silicon area for SRAM within its compute tile. Everything else including Uncore (sub-compute), I/O and back-end advanced packaging should be identical between the two. As such, content value for MTK is also identical for both Humufish and Triggerfish. We estimate its content value is at $13-$14k, 2.5-3x than that for its TPUv8t (Zebrafish). The HVM schedule is 1H28 and the majority of TPUv9 in 2028 will be Humufish, because most of Anthropic's TPUs that year will still be Sunfish or Whalefish provided by AVGO under their current contract. Triggerfish should become more important starting in 2029E and its lifecycle will continue into 2030E.

FIGURE 1: TPUV9 (HUMUFISH & TRIGGERFISH) FLOOR PLAN



Source: Aletheia Capitel/TAG

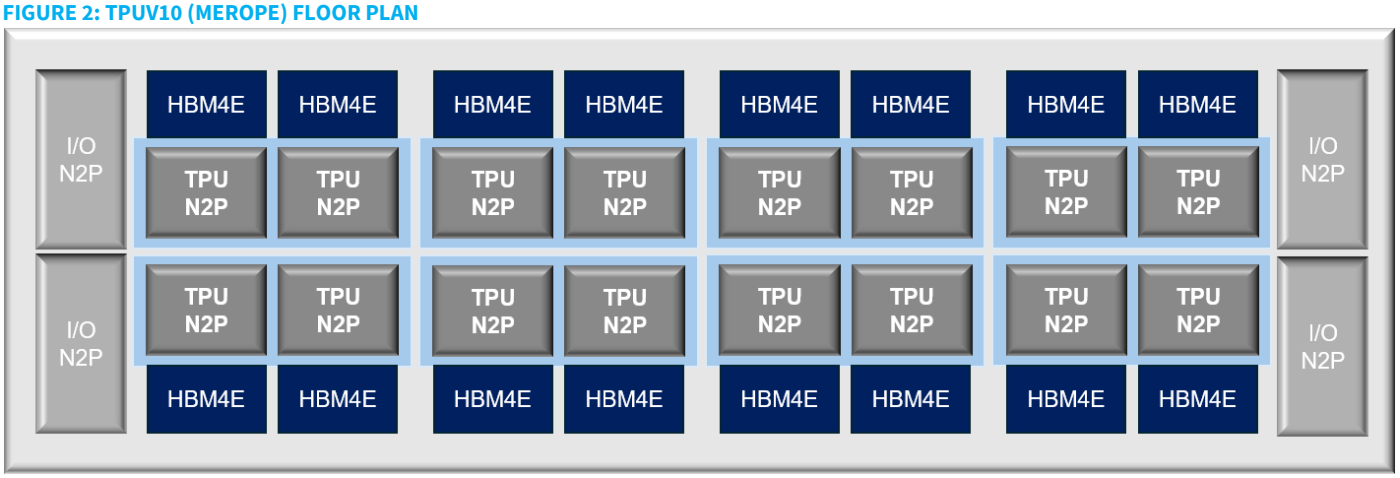
TPUv10 deep dive

Complicity is multiple times than TPUv9

We believe there will be two versions of TPUv10 based on the current plan. The HVM schedule for both is in 2029E so there is still time for any change to the plan. We will update the change accordingly if and when it happens. Our discussions below are all based on our best understanding for the current plan.

The first TPuv10 is called Merope, which we believe is still most likely to be led by AVGO (we will discuss this more in the next section). Merope consists of 16x compute tiles, 8x sub-compute tiles and 4x I/O tiles. Both the compute tiles and I/O tiles are fabricated at TSMC's N2P, while the sub-compute will be powered by TSMC's N3P. Every two compute tiles (top dies, dark grey in the diagram chart) will be stacked on top of one sub-compute (base-die, light blue) leveraging TSMC's SoIC hybrid bonding technology. Merope will also be equipped with 16x HBM4E 8H, with standardized HBM base die under the current plan, for a total of 512GB. All of these tiles are planned to be packaged by Intel's EMIB-T technology.

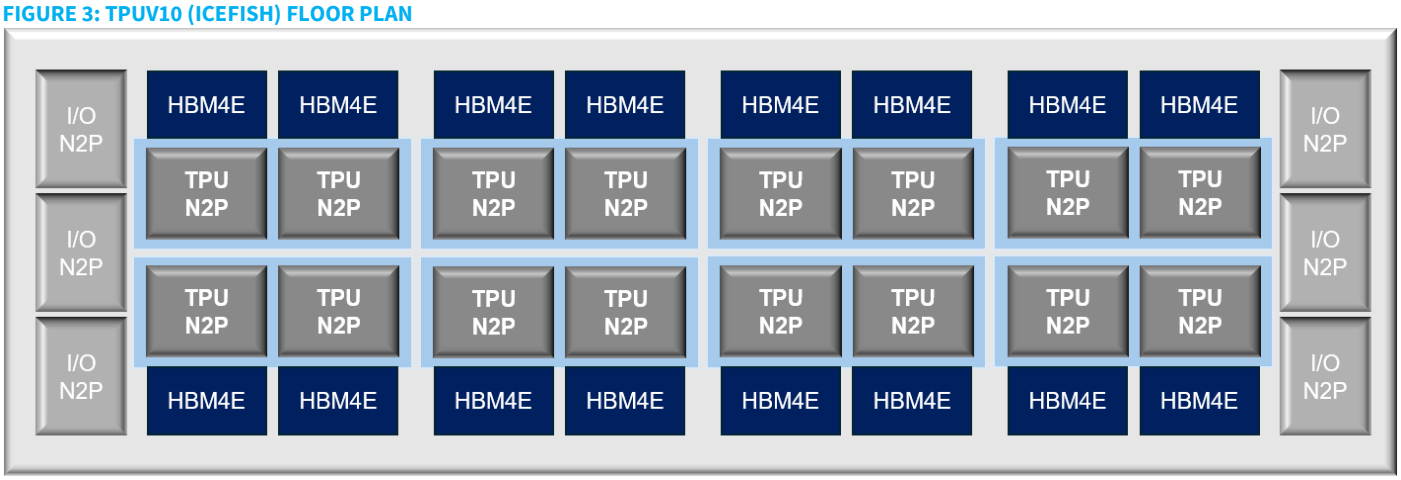
FIGURE 2: TPUV10 (MEROPE) FLOOR PLAN



Source: MediaTek, Aletheia Capital/TAG

The second TPUv10 is called Icefish and we believe MTK will lead this project. It looks a lot like Merope at first glance. It also has 16x compute tiles and 8x sub-compute tiles and every two compute tiles are stacked on top of one sub-compute via TSMC's SoIC hybrid bonding technology. On the other hand, the number of I/O tiles are 6x vs Merope's 4x. In terms of process node, all its logic tiles (compute, sub-compute and I/O) are fabricated by TSMC's N2P. Icefish will also consist of 16x HBM cubes but will be customized HBM4E (HBM4E with customized HBM base die). Icefish will also be packaged by Intel's EMIB-T.

FIGURE 3: TPUV10 (ICEFISH) FLOOR PLAN



Source: MediaTek, Aletheia Capital/TAG

As explained, Merope will have a total of 28x logic tiles and 16x HBM cubes, while Icefish is even more extreme at 30x logic tiles and 16x HBM cubes. This compares with 12x logic tiles and 12x HBM cubes for TPUv9. We also believe TPUv10's packaging size will be at least 1.5x bigger than TPUv9, given the total logic tiles+HBM cubes are 80%-90% more than TPUv9 (46/44 vs 24).

TPUv10: MTK should keep Icefish; Merope details to be confirmed

In addition to the current TPU vendors AVGO and MTK, the market was talking about several new names in recent months, including AMD, MRVL, GUC and Alchip. We believe AMD (and MRVL) are the ones capable of competing in the high value parts, including logic tiles design and advanced packaging. Alchip and GUC are more likely to act in a logistic and placement outsourcing partner role for those chips that Google designs in-house. Despite low margins, these logistic/placement businesses could still contribute meaningful earnings to Alchip or GUC because these businesses are sizeable.

We believe AMD's value proposition is that it can do compute tile design with much lower required margins than AVGO. It is true that AMD may not have comparable I/O Serdes technology vs AVGO or even MTK, but this should not be an issue if it can work together with MRVL. For instance, we believe MRVL has a high chance of supplying AMD's scale-up ASICs starting with MI600 AI GPUs and is likely to provide CPO solutions for the MI500 inter-rack scale-up.

The studies below show how much money Google could save if it moves away from AVGO to other lower-cost suppliers such as AMD or MTK. We analyze the cost and partners' profit for TPUv8i (by MTK) and TPUv8t (by AVGO) and perform a scenario analysis if TPUv8i/v8t is partnered with AMD. In short, AVGO's synthetic GM, if it also prices its TPU in the same way as MTK and potentially AMD, that is, excluding HBM, would be as high as 84% given the dollar profit of $9,000. This compares with MTK's 43% GM and $2,150 dollar profit. Notably, we have assumed a significant higher "other" cost for MTK because the firm needs much more IP licensing vs AVGO. Google's cost per TPUv8t is still less than half of the cost per TPUv8i despite the real cost (excluding design partners' profit) difference between these two being <$2,000 ($7,150 for TPUv8t and $9,000 for TPUv8i).

Then we dive further into the scenario analysis if TPUv8i/TPUv8t is partnered with AMD. Let's start with TPUv8i. Even if we assume that AMD's other cost for all the required IP licensing will be as high as MTK's, Google can still save $5,350 per chip if AMD only requires a 60% GM based on "HBM excluded ASP". The 24% difference in the GM rate (vs AVGO) means more than the 25% difference in cost. Such math should encourage Google to aggressively move from AVGO as long as either its internal team or new design partner (aka AMD) can make up AVGO's design work. On the other hand, moving from MTK to AMD will cost $2,125 more per chip or a 23% increase. Unless Google thinks this extra cost is justified because it can save the internal R&D resources. switching from MTK to AMD doesn't make sense.

The real outcome may not necessarily be binary. For instance, AVGO still has plenty of room to cut prices to secure its share while still keeping its gross margin rate. Google may also have AMD share some of the design work for AVGO's future TPUs as long as Google can integrate work from both firms seamlessly. It is also possible that Google could outsource its compute tile design for MTK's TPU (which Google does internally) to AMD while still asking MTK doing the rest, though we have not heard about this combination. As long as cost-saving is a key consideration for Google, which we believe to be true and exactly why it is pushing COT, the trend favors those lower-cost partners with the knowhow that Google does not have in-house.

FIGURE 4: COST ANALYSIS FOR TPUV8I AND TPUV8T (BASED ON 2027E COST)

Product code	TPU V8t Zebrafish	TPU v8i Sunfish	TPU v8t by AMD Zebrafish	TPU v8i by AMD Sunfish	Remark
Content Analysis					
Logic tile	700	750	700	750	TSMC's wafer
CoWoS	750	950	750	950	20/16 per CoWoS for TPU v8t/TPU v8i
Others	1,400	400	1,400	1,400	MTK/AMD higher for more IP licensing
Total non-Memory cost	2,850	2,100	2,850	3,100	
HBM	4,300	6,900	4,300	6,900	$2.5/Gb, could be higher
Partner's profit	2,150	11,000	4,275	4,650	
GOOGL's cost	**9,300**	**20,000**	**11,425**	**14,650**	
Partner's ASP					
HBM included*	9,300	20,000	11,425	14,650	AVGO prices including HBM
GM	*23%*	***55%***	*37%*	*32%*	
HBM excluded^	**5,000**	13,100	**7,125**	**7,750**	MTK (and AMD) prices excluding HBM
GM	***43%***	*84%*	***60%***	***60%***	
Cost vs existing vendor			*23%*	*-27%*	23% more vs MTK but 27% less vs AVGO

Source: Aletheia Capital/TAG *if ASP by MTK (and AMD) also includes HBM ^if AVGO's ASP exclude HBM

FIGURE 5: TPU DESIGN PARTNERS AND WORKLOAD ALLOCATION, ACTUAL AND POTENTIALS

Product code	TPU V8t Zebrafish	TPU v8i Sunfish	TPU 8.5i Whalefish	TPU v9 Humufish/Triggerfish	TPU v10-M Merope	TPU v10-I Icefish
Compute die	GOOGL	AVGO	AVGO	GOOGL	GOOGL/AMD/AVGO	GOOGL
Sub-compute/Base die	N.A.	N.A.	N.A.	MTK	AVGO	GOOGL
I/O	MTK	AVGO	AVGO	MTK	AVGO	MTK
SoIC 3D Packaging	N.A.	N.A.	N.A.	N.A.	AVGO/AMD	GOOGL/GUC
2.5D Packaging	CoWoS	CoWoS	CoWoS	EMIB-T	EMIB-T	EMIB-T
HBM Procurement	GOOGL	AVGO	AVGO	GOOGL	GOOGL	GOOGL
HVM	4Q26	4Q26	4Q27	1H28	1H29	2029

Source: MediaTek, Aletheia Capital/TAG

Estimated content value for MTK

Assuming MTK will still gain TPUv10 Icefish, we estimate its content value will see at least 50%-80% generation-to-generation increase from its TPUv9.

For TPUv9, MTK is in charge of a total of 8x logic tiles, including 4x sub-compute and 4x I/O. All are powered by TSMC's N3P. In addition, MTK is taking care of the EMIB-T order placement (to Intel) and material sourcing. Including other IP licensing fees, and so on, we estimate total content per chip is staying at $13-$14k. Notably, MTK is not taking care of the compute tile design and wafer placement and therefore either it is Humufish or Triggerfish that does not impact MTK's content value.

Moving on to TPUv10, Google will take care of more tiles compared with TPUv9. We believe it will self-handle the compute tiles (total 16x), sub-compute or the base tiles (total 8x), and the 3DIC packaging of these two. As for MTK (or any other design partner if MTK does not win the project), it will be in charge of 6x I/O (powered by N2P), 2.5D packaging, and possibly the 16x HBM4E base die. The 2.5D back-end cost should grow by more than 50% generation-to-generation because the package size will already grow by 50% from TPUv9 to TPUv10 Icefish, and back-end cost growth should outgrow the size growth because of yield control difficulties. As for the front-end logic tiles, it is true that MTK will own fewer logic tiles (from 8x to 6x), but it will be more than offset by geometry migration from N3P to N2P.Combining all of these together, we believe MTK's content will approach $20k or ~50% generation-to-generation growth. If MTK is also in charge of the HBM base die, MTK's content should be more than $20K and it would not be impossible for it to approach $25k.

Earnings estimates and stock recommendation

Earnings revision

We tweaked our 2026E earnings down by 7% as we are revising down our core business estimates in 4Q26E. We now assume 4Q26E core business will slide by ~10% QoQ driven mainly by a seasonal pattern of smart-edge devices and customers' pull-in ahead of MTK's price increase in September. On the other hand, we are revising up our 2027/28E earnings by 6%/28% driven mainly by our estimates for higher ASIC revenue. Notably, we believe MTK not only has TPU but also has Bytedance's ASIC business starting in 2027E, with a revenue contribution of $1b-$2b/$2b-3b for 2027/28E. We believe ASIC business for Bytedance is similar to QCOM's, that MTK is basically a capacity supplier and chips are still based on Bytedance's design. Our revised 2026/27/28E EPS are 6%/30%/18% ahead of Factset consensus estimates.

Reiterate BUY with TP of TWD6,500

We reiterate our BUY rating for MTK and raise our TP to TWD6,500 from the previous TWD5,300, based on unchanged 25x average 2027/28E P/E, which also implies 20x 2028E P/E. We think MTK deserves a higher P/E vs its AI compute peers globally, justified by its rapid earnings growth in the next 24 months (+155%/+75% in 2027/28E) thanks to its market share gains and better earnings predictability. We may be able to rollover to later years' multiples once we can identify more ASIC project wins and quantify them. In a bull case scenario —MTK not only secures TPUv10 but also wins another few ASIC projects it is working on now —it is not impossible to see TWD400-500 earnings power in outer years and this can further drive the shares – we just need more time for the company to prove this. Near-term catalysts include TPUv8t HVM (4Q26), TPUv10 design partner confirmation (in the next few months), a new ASIC customer (Bytedance), and the TPUv9 tapeout result (tapeout in 4Q26E and result in 1Q27E).

FIGURE 6: ALETHEIA'S EARNINGS REVISION FOR MEDIATEK

	3Q26E			4Q26E			2026E			2027E			2028E		
(NT$bn)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)
Sales	156.7	147.6	6.1	221.8	239.3	(7.3)	679.8	688.3	(1.2)	1,360.3	1,239.3	9.8	2,096.5	1,707.2	22.8
Gross profit	72.2	68.5	5.5	98.9	107.3	(7.8)	310.5	315.4	(1.6)	597.5	549.2	8.8	909.5	748.3	21.5
Operating profit	23.4	22.3	5.0	43.3	57.1	(24.2)	112.5	126.7	(11.2)	316.0	307.8	2.7	566.6	453.9	24.8
Net profit	25.1	23.2	8.2	42.6	52.8	(19.4)	116.2	125.1	(7.1)	296.0	278.7	6.2	516.5	402.8	28.2
EPS (NT$)	15.75	14.60	7.9	26.73	33.10	(19.2)	72.92	78.60	(7.2)	185.78	174.90	6.2	324.22	252.90	28.2
Gross margins (%)	46.1	46.4	(0.3)	44.6	44.8	(0.2)	45.7	45.8	(0.1)	43.9	44.3	(0.4)	43.4	43.8	(0.5)
OP margins (%)	15.0	15.1	(0.2)	19.5	23.9	(4.3)	16.5	18.4	(1.9)	23.2	24.8	(1.6)	27.0	26.6	0.4
Net margins (%)	16.0	15.7	0.3	19.2	22.1	(2.9)	17.1	18.2	(1.1)	21.8	22.5	(0.7)	24.6	23.6	1.0

Source: MediaTek, Aletheia Capital/TAG

FIGURE 7: ALETHEIA'S EARNINGS ESTIMATES FOR MEDIATEK VS CONSENSUS

	3Q26E			4Q26E			2026E			2027E			2028E		
(NT$bn)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)
Sales	156.7	156.8	(0.1)	221.8	208.6	6.3	679.8	661.7	2.7	1,360.3	1,133.4	20.0	2,096.5	1,975.3	6.1
Gross profit	72.2	72.1	0.3	98.9	93.3	6.0	310.5	303.6	2.3	597.5	499.6	19.6	909.5	801.3	13.5
Operating profit	23.4	24.2	(3.1)	43.3	38.1	13.7	112.5	107.9	4.3	316.0	255.2	23.8	566.6	517.9	9.4
Net profit	25.1	24.7	1.5	42.6	36.5	16.6	116.2	109.5	6.1	296.0	231.2	28.0	516.5	453.1	14.0
EPS (NT$)	15.75	15.45	1.9	26.73	23.03	16.1	72.92	68.60	6.3	185.78	143.23	29.7	324.22	277.41	16.9
Gross margins (%)	46.1	46.0	0.1	44.6	44.7	(0.1)	45.7	45.9	(0.2)	43.9	44.1	(0.2)	43.4	40.6	2.8
OP margins (%)	15.0	15.4	(0.5)	19.5	18.3	1.3	16.5	16.3	0.2	23.2	22.5	0.7	27.0	26.2	0.8
Net margins (%)	16.0	15.8	0.2	19.2	17.5	1.7	17.1	16.5	0.5	21.8	20.4	1.4	24.6	22.9	1.7

Source: FactSet, MediaTek, Aletheia Capital /TAG

Earnings model

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Shipments (m units)																
Smartphones	127	102	101	111	95	80	74	71	78	79	82	83	441	320	322	328
5G Smartphones	*57*	*54*	*53*	*56*	*49*	*41*	*39*	*36*	*38*	*39*	*42*	*43*	220	165	162	168
4G LTE Smartphones	*71*	*48*	*48*	*55*	*46*	*39*	*35*	*35*	*40*	*40*	*40*	*40*	222	155	160	160
Tablets/Chromebooks	5	5	6	5	5	5	4	4	4	4	4	4	21	17	16	16
D-TVs	40	44	40	36	41	44	42	30	34	37	41	39	160	157	152	152
AI ASICs	-	-	-	-	-	-	-	1	0.8	0.9	1.1	1	-	1	4	6
Sales mix (%)																
Smartphones	56	52	53	59	49	41	39	29	23	20	18	19	55	38	20	14
Tablets/Chromebooks	1	1	2	1	1	2	1	1	1	1	1	0	1	1	1	0
D-TVs	11	12	11	10	14	17	16	8	8	8	7	6	11	13	7	5
AI ASICs	-	-	-	-	-	-	-	36	46	49	53	58	-	12	52	66
Others	31	35	34	30	36	41	44	27	23	22	20	16	32	36	20	15

P&L model (NT$bn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
TWD/USD	*32.5*	*30.8*	*29.9*	*31.0*	*31.5*	*32.0*	*31.8*	*31.8*	*31.8*	*31.8*	*31.8*	*31.8*	*31.1*	*31.8*	*31.8*	*31.8*
Revenue	**153.3**	**150.4**	**142.1**	**150.2**	**149.2**	**152.2**	**156.7**	**221.8**	**279.3**	**307.3**	**353.5**	**420.3**	**596.0**	**679.8**	**1,360.3**	**2,096.5**
COGS	(73.9)	(70.7)	(70.2)	(75.1)	(74.3)	(81.9)	(84.4)	(122.9)	(155.8)	(171.8)	(198.2)	(237.0)	(289.9)	(363.4)	(762.8)	(1,187.0)
Gross profit	73.8	73.9	66.1	69.3	69.1	70.3	72.2	98.9	123.5	135.5	155.2	183.3	283.1	310.5	597.5	909.5
OPEX	(43.8)	(44.5)	(43.9)	(47.4)	(46.2)	(47.4)	(48.8)	(55.6)	(67.6)	(69.0)	(70.9)	(74.0)	(179.6)	(198.0)	(281.5)	(342.9)
Operating profit	30.1	29.4	22.2	21.8	22.9	22.9	23.4	43.3	55.9	66.5	84.3	109.3	103.5	112.5	316.0	566.6
Non-op	4.5	3.8	7.8	5.3	4.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1	21.4	19.4	20.4	20.4
Profit before tax	34.6	33.2	30.0	27.1	27.0	28.0	28.5	48.4	61.0	71.6	89.4	114.4	124.9	131.9	336.3	587.0
Tax	(5.0)	(5.2)	(4.5)	(4.1)	(2.6)	(3.4)	(3.4)	(5.8)	(7.3)	(8.6)	(10.7)	(13.7)	(18.8)	(15.2)	(40.4)	(70.5)
Net profit	29.3	27.8	25.2	22.9	24.2	24.3	25.1	42.6	53.7	63.0	78.7	100.6	105.3	116.2	296.0	516.5
FD EPS (NT$)	***18.4***	***17.5***	***15.8***	***14.4***	***15.2***	***15.3***	***15.8***	***26.7***	***33.7***	***39.5***	***49.4***	***63.2***	***66.1***	***72.9***	***185.8***	***324.2***

Margins (%)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
GM	48.1	49.1	46.5	46.1	46.3	46.2	46.1	44.6	44.2	44.1	43.9	43.6	47.5	45.7	43.9	43.4
OPM	19.6	19.5	15.6	14.5	15.3	15.0	15.0	19.5	20.0	21.6	23.8	26.0	17.4	16.5	23.2	27.0
PTM	22.5	22.1	21.1	18.1	18.1	18.4	18.2	21.8	21.8	23.3	25.3	27.2	21.0	19.4	24.7	28.0
NP	19.1	18.5	17.7	15.3	16.2	16.0	16.0	19.2	19.2	20.5	22.3	23.9	17.7	17.1	21.8	24.6

QoQ chg (%)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Revenue	11.1	(1.9)	(5.5)	5.7	(0.7)	2.0	2.9	41.6	25.9	10.0	15.0	18.9	-	-	-	-
Gross profit	10.2	0.1	(10.5)	4.8	(0.3)	1.8	2.7	36.9	24.9	9.7	14.6	18.1	-	-	-	-
Operating profit	40.4	(2.2)	(24.4)	(1.6)	4.8	(0.1)	2.4	84.9	29.1	19.0	26.8	29.6	-	-	-	-
Net profit	23.3	(5.0)	(9.4)	(9.1)	5.4	0.7	3.1	69.7	26.0	17.4	24.9	27.9	-	-	-	-
EPS	23.5	(5.2)	(9.4)	(9.1)	5.4	0.7	3.1	69.7	26.0	17.4	24.9	27.9	-	-	-	-

YoY chg (%)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Revenue	15	18	8	8.8	(3)	1	10	47.7	87	102	126	89.5	12	14	100	54
Gross profit	6	19	3	3.4	(6)	(5)	9	42.7	79	93	115	85.4	7	10	92	52
Operating profit	(7)	18	(7)	2.0	(24)	(22)	6	98.3	144	191	260	152.4	1	9	181	79
Net profit	(7)	8	(0)	(3.6)	(18)	(13)	(1)	85.7	122	159	214	136.3	(1)	10	155	75
EPS	(7)	8	(1)	(3.6)	(18)	(13)	(0)	85.8	122	159	214	136.3	(1)	10	155	75

Source: Company, Aletheia Capital/TAG

Financial Statements

INCOME STATEMENT

FYE Dec (TWDb)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	433	531	596	680	1,360	2,096
COGS	(208)	(246)	(290)	(363)	(763)	(1,187)
Gross profit	207	263	283	311	598	909
Operating expense	(136)	(161)	(180)	(198)	(282)	(343)
EBITDA	90	123	126	136	339	590
Operating profit	72	102	103	112	316	567
Net non-operating income	15	17	21	19	20	20
Other income (expense)	0	0	0	0	0	0
Profit before tax	87	120	125	132	336	587
Tax	(10)	(12)	(19)	(15)	(40)	(70)
Net profit	77	107	106	117	296	517
EPS (TWD)	48.33	66.78	66.11	72.92	185.78	324.22
Change YoY						
Revenue	-21%	22%	12%	14%	100%	54%
Gross profit	-23%	27%	7%	10%	92%	52%
Operating profit	-43%	43%	1%	9%	181%	79%
Net profit	-35%	39%	-1%	10%	154%	75%
EPS	-35%	38%	-1%	10%	155%	75%

BALANCE SHEET

FYE Dec (TWDb)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Cash/ST investments	181	220	248	215	279	484
Inventory	43	58	67	112	233	295
Receivable	55	44	62	101	196	254
Current assets	291	351	397	451	731	1,056
Fixed assets	53	57	60	64	58	53
Goodwill & intangibles	81	82	79	98	98	98
Other LT assets	210	208	206	226	226	226
LT assets	344	347	346	387	382	377
Total assets	635	698	744	838	1,113	1,433
Payable	39	41	49	73	145	186
ST debt	3	2	1	17	17	17
Current liabilities	232	267	303	334	406	447
LT debt	29	26	31	34	34	34
Total liabilities	261	293	335	369	441	481
Retained earnings	213	211	220	254	454	728
Equity	374	405	409	445	648	927
Liability&Equity	635	698	744	822	1,097	1,417

CASH FLOW

FYE Dec (TWDb)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Net profit	77	106	105	116	296	517
Dep. & amortization	18	21	23	23	23	23
Chg in WC	30	(2)	(19)	(59)	(144)	(80)
Others	40	31	54	(10)	0	0
OPN cash flow	166	156	163	70	175	460
Capex	(9)	(14)	(15)	(19)	(18)	(18)
Chg in investment	(12)	(14)	(11)	4	0	0
Others	(3)	(8)	(12)	(16)	0	0
Investing CF	(29)	(36)	(38)	(32)	(18)	(18)
Chg in debt	(3)	(2)	0	16	0	0
Equity raised	0	0	0	0	0	0
Dividend paid	(121)	(88)	(86)	(85)	(93)	(237)
Others	5	2	(2)	(1)	0	0
Financing CF	(119)	(90)	(88)	(70)	(93)	(237)
FX & other adj	(1)	8	(6)	2	0	0
Chg in cash	18	38	32	(30)	64	205
Beginning cash	148	165	204	235	205	269
End cash	165	204	235	205	269	474
FCF	157	142	148	51	157	442

RATIO & PER SHARE DATA

FYE Dec (TWD)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Gross margin	48%	50%	47%	46%	44%	43%
EBITDA margin	21%	23%	21%	20%	25%	28%
Op. Profit margin	17%	19%	17%	17%	23%	27%
Net Profit margin	18%	20%	18%	17%	22%	25%
ROE (%)	19%	27%	26%	27%	54%	66%
ROA (%)	12%	16%	15%	15%	30%	41%
Gross gearing	9%	7%	8%	12%	8%	6%
Net gearing	-47%	-54%	-60%	-44%	-40%	-50%
Asset turnover (%)	70%	80%	83%	86%	139%	165%
Leverage (x)	1.5	1.7	1.8	1.9	1.8	1.6
Payable days	52.9	59.1	56.3	61.2	52.3	51.1
Receivable days	40.4	34.2	32.5	43.7	39.8	39.3
Inventory days	100.0	75.5	79.1	90.2	82.6	81.4
EPS (TWD)	48.3	66.8	66.1	72.9	185.8	324.2
BVPS	234.4	254.0	256.8	279.4	406.6	582.2
Net cash per share	111.3	136.5	154.8	124.0	164.2	292.8
SPS	271.5	332.7	374.0	426.7	853.9	1,316.0
FCFPS	98.2	89.2	92.7	31.8	98.8	277.3
Total returns (%)	1.5	1.1	1.1	1.1	1.2	3.0

Source: Company, Aletheia Capital /TAG

1 October 2026
Technology Hardware
USA

Nvidia

↔ Buy

#Semiconductor #DataCenter #AI

Price: **$ 228.38** Target Price: **$ 340** Upside: **49%** Total return: **52%** Risk: High

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Nvidia Corp.

Bloomberg	NVDA US
RIC	NVDA.N
Market cap (USDbn)	5,516
Average daily T/O (USD|bn)	29.1

Near-Term Update

We reiterate BUY on NVDA and a TP of $340, based on 20x FY28E (CY27E) P/E. Punished by its market cap and its unit share loss to TPUs, the shares were only up by ~20% YTD and underperformed ~70% for SOX massively. We are seeing several dynamics that favour NVDA. Google's TPU is the only ASIC with volume momentum. All other CSPs are buying more of NVDA's GPUs with lowered or delayed ASICs, such as AWS and SpaceX. Token consumption by open-source models is growing at a much faster rate than frontier models and they are all running on NVDA GPUs. Lastly, NVDA's GPU units/revenue per GW could still see upside. Its DSX platform (MaxQ and MaxLPS) could sufficiently increase its GPU unit per GW by 20-40% compared with the TDP design, and customers can also reduce token costs. Higher volume potential could offset its ASP pressure driven by a memory de-bundle (starting from LPDDR in 2027E) and continue to drive revenue growth. In the near-term, the recent Rubin ramp may see some adjustment because of HBM issues, but the impact is manageable. Its new $150b share buyback announcement, lifting the total unused quota to $235b before end-FY28, should provide additional support to the shares.

Rubin production on schedule; some volume adjustment due to HBM

Rubin HVM is on schedule to ramp in C3Q26, but we anticipate some volume shortfall because of HBM issues from Hynix. This leads FT at KYEC may only be fully ramped in November vs the previous expectation of September/October. Rubin powered by Hynix's HBM has resumed its CoWoS production, but we believe the volume is yet to reach the target level. As such, the current target production volume of 2.2m Rubin in 2026E could be proven challenging. We also tweak down our FY27E Rubin shipments of 1.6m to 1.5m accordingly. The impact on our earnings is negligible.

Favorable industry dynamics

It is always hard to capture global total token consumption and market share by model, because token consumption for those frontier models via consumer-facing apps or enterprise API is not publicly available. However, statistics on some websites, such as Open Router, could be still used as a reference. Collective token consumption market share by the top three frontier model providers (Google, OAI and Anthropic) was 60%, but the numbers have shrunk to 39% now, with most of the decline coming from Google's Gemini. All of those open-source models, except those for China domestic-only, are powered by NVDA's GPUs. Should token consumption by open-source models continue to grow their weightings, we could see NVDA further grow its market share from its already dominant position. To gain exposure to open-source models could be one important reason why AMD announced in recent days that it would buy World Labs for $8.2b.

KEY FINANCIAL AND VALUATION METRICS

FYE Jan (US$ mn, non-GAAP)	FY22A	FY23A	FY24A	FY25A	FY26A	FY27E	FY28E
Revenue	26,914	26,974	60,922	130,497	215,938	407,605	758,364
OPN Profit	12,691	9,039	37,134	86,788	135,271	267,713	484,306
Net Profit	11,258	8,366	32,312	74,266	115,887	226,018	408,008
FD GAAP EPS ($)	0.38	0.17	1.19	2.94	4.90	9.95	16.74
FD Non-GAAP EPS ($)	***0.44***	***0.33***	***1.30***	***2.99***	***4.86***	***9.32***	***17.00***
Consensus Non-GAAP EPS ($)	-	-	-	-	-	9.27	15.74
PER (X) - Non-GAAP	514.3	685.8	176.1	76.3	47.0	24.5	13.4
PBR (X)	217.5	259.1	132.5	71.4	35.6	17.3	8.5
EV/EBITDA	512.2	1002.4	164.2	67.2	39.4	18.7	10.9
ROE (%)	44.8	17.9	91.5	119.2	101.5	101.3	83.6
ROA (%)	26.7	10.2	55.7	82.2	75.4	77.3	68.3

Source: NVDA, FactSet, Aletheia/TAG

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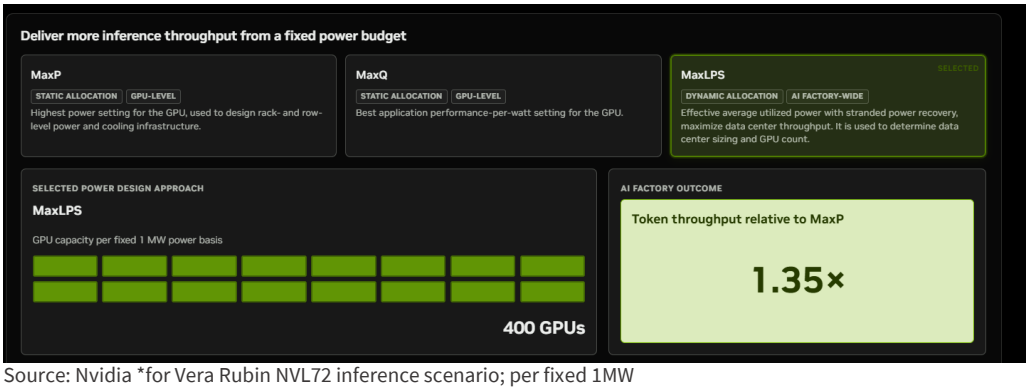
NVDA's DSX platform: MaxP, MaxQ, and MaxLPS

NVDA's DSX is its reference design for AI factories starting from the GB200 NVL72. It covers from platform management software to hardware rack installation layout. The key idea is that not all GPUs will run at their peak power all the time, with NVDA identifying three different power settings: MaxP (Max Performance), MaxQ (Max Efficiency), and MaxLPS (Max Land Power Shell). The former two are static power settings at the GPU level while the last refers to dynamic power allocation at the AI factory level (which will also lead to the lowest power consumption at GPU/rack level).

Without going into detail, NVDA claims that MaxQ and MaxLPS status can generate the best token cost for most applications. Even training can be done most efficiently under MaxQ rather than MaxP. Given GPUs are running at much lower power compared with MaxP, which is running at its TDP (Thermal Designed Power), every 1GW of infrastructure will be able to house more GPUs if the data center is designed under MaxQ or MaxLPS. Every 1GW can accommodate 275k Rubin GPUs under Vera-Rubin NVL72 architecture at a PUE of 1.1 (38.2k racks). The number of GPUs per 1GW will increase to 330k/400k under MaxQ and MaxLPS design, representing 20%/46% more GPUs per 1GW. NVDA disclosed details here.

NVDA will continue this DSX offering and we already have some initial color for the Rubin Ultra generation. We noted that rack power for the Rubin Ultra NVL72 is 235kW under a MaxP configuration, which is just a small increase from 225kW for Vera Rubin NVL72. Under MaxQ and MaxLPS, rack power will go down to ~170kW and ~140kW. For the GB300 NVL72, MaxP rack power is 150kW and MaxLPS is about 100kW. This suggests the Rubin Ultra NVL72 could consume lower power vs the GB300 if the former runs at MaxLPS status vs MaxP for the latter. Also, DC operators can freely switch among these modes by software control, which provides DC operators the flexibility to run the AI factory at the most efficient mode for different demands. The bottom line is – DCs operator could install more GPUs vs the theoretical number of GPUs defined by TDP, and such "over installation" can be as much as 45% in the case of NVDA's Vera Rubin NVL72.

FIGURE 1: NVDA'S MAXLPS ALLOWS 35% MORE GPU CAPACITY VS WHAT TDP SUGGESTED



Source: Nvidia *for Vera Rubin NVL72 inference scenario; per fixed 1MW

FIGURE 2: NVDA - ESTIMATE REVISION

	3Q27E			4Q27E			FY27E			FY28E		
(US$ bn)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)
Sales	110	111	(0.6)	120	121	(1.3)	408	408	(0.0)	758	755	0.4
Gross profit	81	82	(0.7)	89	90	(1.4)	303	304	(0.0)	550	548	0.4
OP profit	72	72	(0.4)	78	79	(1.2)	268	267	0.3	484	483	0.3
Net profit	60	61	(0.7)	66	67	(1.4)	226	226	0.1	408	407	0.2
EPS (US$)	2.50	2.51	(0.6)	2.74	2.77	(1.2)	9.32	9.31	0.1	17.00	16.97	0.2
Gross margin (%)	74.0	74.0	(0.1)	74.1	74.1	(0.1)	74.5	74.5	(0.0)	72.5	72.5	(0.0)
OP margin (%)	65.1	65.0	0.1	65.4	65.3	0.1	65.7	65.5	0.2	63.9	63.9	(0.0)
Net margin (%)	54.9	55.0	(0.0)	55.2	55.2	(0.0)	55.5	55.4	0.1	53.8	53.9	(0.1)

Source: NVDA, Aletheia/TAG

FIGURE 3: NVDA - ESTIMATE VS CONSENSUS

	3Q27E			4Q27E			FY27E			FY28E		
(US$ bn)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)
Sales	110	109	1.2	120	124	(3.6)	408	409	(0.4)	758	691	9.7
Gross profit	81	81	0.7	89	89	(0.9)	303	303	0.1	550	504	9.1
OP profit	72	72	(0.2)	78	79	(1.3)	268	268	(0.2)	484	455	6.4
Net profit	60	60	1.1	66	66	(0.4)	226	226	0.2	408	380	7.3
EPS (US$)	2.50	2.47	1.1	2.74	2.73	0.2	9.32	9.27	0.6	17.00	15.74	8.0
Gross margin (%)	74.0	75.0	(1.0)	74.1	74.8	(0.8)	74.5	74.9	(0.4)	72.5	74.3	(1.9)
OP margin (%)	65.1	66.0	(0.9)	65.4	63.9	1.5	65.7	65.5	0.1	63.9	65.9	(2.0)
Net margin (%)	54.9	55.0	(0.1)	55.2	53.4	1.8	55.5	55.1	0.3	53.8	55.0	(1.2)

Source: NVDA, Aletheia/TAG

Earnings Model

FYE Jan	Apr-25	Jul-25	Oct-25	Jan-26	Apr-26	Jul-26	Oct-26	Jan-27	Jan-25	Jan-26	Jan-27	Jan-28
Sales mix, US$ mn	**FY1Q26**	**2Q26**	**3Q26**	**4Q26**	**FY1Q27**	**2Q27**	**3Q27E**	**4Q27E**	**FY25A**	**FY26A**	**FY27E**	**FY28E**
Data center	**39,112**	**41,096**	**51,215**	**62,314**	**75,246**	**89,023**	**102,209**	**111,956**	**115,186**	**193,737**	**378,434**	**727,349**
-HGX (A/H/B/R)	17,635	10,788	12,011	10,334	12,686	18,439	13,280	8,200	92,453	50,768	52,605	50,000
-Rack module	16,040	22,400	31,017	41,000	47,560	53,660	67,607	78,878	4,200	110,457	247,705	510,000
-Networking	4,957	7,252	8,187	10,980	15,000	16,924	21,323	24,878	12,990	31,376	78,124	160,849
-Others	480	656	0	0	0	0	0	0	5,543	1,136	0	6,500
FYE Jan	Apr-26	Jul-26	Oct-26	Jan-26	Apr-26	Jul-26	Oct-26	Jan-27	Jan-25	Jan-26	Jan-27	Jan-28
Gaming	3,763	4,287	4,265	3,727	3,537	3,517	4,003	3,798	11,350	16,042	14,854	15,261
Professional Visualization	509	601	760	1,321	1,915	2,682	2,816	2,956	1,878	3,191	10,369	10,831
Automotive	567	586	592	604	737	762	770	785	1,694	2,349	3,054	3,970
OEM and IP and other	111	173	174	161	180	238	238	238	389	619	894	952
Total	**44,062**	**46,743**	**57,006**	**68,127**	**81,615**	**96,221**	**110,036**	**119,734**	**130,497**	**215,938**	**407,605**	**758,364**
Data center	89%	88%	90%	91%	92%	93%	93%	94%	88%	90%	93%	96%
Gaming	9%	9%	7%	5%	4%	4%	4%	3%	9%	7%	4%	2%
Professional Visualization	1%	1%	1%	2%	2%	3%	3%	2%	1%	1%	3%	1%
Automotive	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%
OEM and IP and other	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
P&L US$ mn, non-GAAP	**FY1Q26**	**2Q26**	**3Q26**	**4Q26**	**FY1Q27**	**2Q27**	**3Q27E**	**4Q27E**	**FY25A**	**FY26A**	**FY27E**	**FY28E**
Revenue	**44,062**	**46,743**	**57,006**	**68,127**	**81,615**	**96,221**	**110,036**	**119,734**	**130,497**	**215,938**	**407,605**	**758,364**
COGS	12,730	13,021	15,109	16,987	20,383	24,033	28,662	31,062	31,992	57,847	104,140	208,684
Gross profit	31,332	33,722	41,897	51,140	61,232	72,188	81,374	88,672	98,505	158,091	303,465	549,681
OPEX	4,993	5,361	5,800	6,666	7,449	8,232	9,754	10,317	11,717	22,820	35,753	65,374
- R&D, GAAP	3,989	4,291	4,705	5,512	6,321	7,054	8,253	8,741	12,914	18,497	30,368	53,844
- SGA, GAAP	1,041	1,122	1,134	1,282	1,300	1,354	1,548	1,685	3,491	4,579	5,887	12,134
- Non-GAAP reconciliaiton	(37)	(52)	(39)	(128)	(172)	(176)	(47)	(108)	(4,688)	(256)	(503)	(603)
OP profit	26,339	28,361	36,097	44,474	53,783	63,956	71,620	78,354	86,788	135,271	267,713	484,306
Non-OP	448	520	573	621	457	300	346	353	1,550	2,162	1,456	1,607
PTI	26,787	28,881	36,670	45,095	54,240	64,256	71,966	78,707	88,338	137,433	269,169	485,913
Tax	(3,952)	(4,910)	(6,558)	(6,126)	(8,692)	(10,302)	(11,538)	(12,619)	(14,072)	(21,546)	(43,151)	(77,905)
Net profit	22,835	23,971	30,112	38,969	45,548	53,954	60,428	66,088	74,266	115,887	226,018	408,008
FD EPS, non-GAAP	**0.93**	**1.04**	**1.30**	**1.59**	**1.87**	**2.22**	**2.50**	**2.74**	**2.99**	**4.86**	**9.32**	**17.00**
FD EPS, GAAP	0.76	1.08	1.30	1.76	2.39	2.46	2.43	2.67	2.94	4.90	9.95	16.74
Profitability	**FY1Q26**	**2Q26**	**3Q26**	**4Q26**	**FY1Q27**	**2Q27**	**3Q27E**	**4Q27E**	**FY25A**	**FY26A**	**FY27E**	**FY28E**
Gross margin	71.1%	72.1%	73.5%	75.1%	75.0%	75.0%	74.0%	74.1%	75.5%	73.2%	74.5%	72.5%
OPN margin	59.8%	60.7%	63.3%	65.3%	65.9%	66.5%	65.1%	65.4%	66.5%	62.6%	65.7%	63.9%
PTI margin	60.8%	61.8%	64.3%	66.2%	66.5%	66.8%	65.4%	65.7%	67.7%	63.6%	66.0%	64.1%
Net margin	51.8%	51.3%	52.8%	57.2%	55.8%	56.1%	54.9%	55.2%	56.9%	53.7%	55.5%	53.8%
YoY chg.	**FY1Q26**	**2Q26**	**3Q26**	**4Q26**	**FY1Q27**	**2Q27**	**3Q27E**	**4Q27E**	**FY25A**	**FY26A**	**FY27E**	**FY28E**
Revenue	69%	56%	62%	73%	85%	106%	93%	76%	114%	65%	89%	86%
Gross profit	52%	48%	59%	77%	95%	114%	94%	73%	119%	60%	92%	81%
OPN profit	46%	42%	55%	74%	104%	126%	98%	76%	134%	56%	98%	81%
PTI	46%	42%	55%	74%	102%	122%	96%	75%	134%	56%	96%	81%
Net profit	50%	41%	50%	77%	99%	125%	101%	70%	130%	56%	95%	81%
FD EPS	52%	53%	60%	79%	100%	113%	92%	72%	131%	63%	92%	82%

Source: NVDA, Aletheia/TAG

Financial Statements

INCOME STATEMENT

FYE Jan (US$ mn)	FY24A	FY25A	FY26A	FY27E	FY28E
Revenue	60,922	130,497	215,938	407,605	758,364
COGS	16,621	32,638	62,475	104,362	208,887
Gross profit	44,301	97,859	153,463	303,244	549,477
Operating expense	11,329	16,405	23,076	36,255	65,978
EBITDA	34,718	84,351	142,252	295,581	493,519
Operating profit	32,972	81,454	130,387	266,988	483,499
Interest income (expense)	608	1,539	2,041	1,245	1,076
Other income (expense)	238	1,033	9,022	23,778	1,138
Profit before tax	33,818	84,026	141,450	292,011	485,713
Tax	(4,059)	(11,146)	(21,383)	(50,592)	(83,973)
Net profit	29,759	72,880	120,067	241,419	401,740
EPS US$, GAAP	1.19	2.94	4.90	9.95	16.74
EPS US$, Non-GAAP	1.30	2.99	4.86	9.32	17.00
Change YoY					
Revenue	126%	114%	65%	89%	86%
Gross profit	188%	121%	57%	98%	81%
Operating profit	681%	147%	60%	105%	81%
Net profit	581%	145%	65%	101%	66%
EPS	289%	131%	63%	92%	82%

BALANCE SHEET

FYE Jan (US$ mn)	FY24A	FY25A	FY26A	FY27E	FY28E
Cash/ST investment	25,984	43,210	62,556	147,362	258,132
Inventory	5,282	10,080	21,403	51,854	165,713
Receivable	9,999	23,065	38,466	79,822	162,514
Current assets	44,345	80,126	125,605	283,163	592,398
Fixed assets	5,260	8,076	13,250	22,773	32,998
Goodwill & intangibles	5,542	5,995	24,138	24,726	26,718
Other LT assets	10,581	17,404	43,810	87,166	105,951
LT assets	21,383	31,475	81,198	134,665	165,666
Total assets	65,728	111,601	206,803	417,829	758,064
Payable	2,699	6,310	9,812	23,046	42,963
Other liability	7,932	11,737	22,351	27,960	27,960
Current liabilities	10,631	18,047	32,163	51,006	70,923
LT liabilities	12,119	14,227	17,347	47,269	45,269
Total liabilities	22,750	32,274	49,510	98,275	116,192
Common shares & capital	13,134	11,261	10,142	26,652	60,252
Retained earnings	29,817	68,038	146,973	292,926	581,646
Equity	42,978	79,327	157,293	319,553	641,873
Liability&Equity	65,728	111,601	206,803	417,829	758,064

CASH FLOW

FYE Jan (US$ mn)	FY24A	FY25A	FY26A	FY27E	FY28E
Net profit	29,759	72,880	120,067	241,419	401,740
Dep. & amortization	1,508	1,864	2,843	4,815	8,882
Chg in WC	(4,516)	(9,889)	(13,016)	(53,910)	(178,548)
Others	1,339	(766)	(7,176)	(29,677)	(18,785)
OPN cash flow	28,090	64,089	102,718	162,647	213,289
Capex	(1,069)	(3,236)	(6,042)	(10,826)	(21,099)
Chg in investment	(83)	(1,007)	(1,535)	(298)	0
Others	(985)	(1,293)	(30,418)	(21,792)	0
Investing CF	(10,566)	(20,421)	(52,228)	(51,516)	(41,099)
Chg in debt	(1,250)	(1,250)	0	23,896	(2,000)
Equity raised/Repurchased	(9,130)	(33,216)	(39,442)	(76,129)	(85,500)
Dividend paid	(395)	(834)	(974)	(18,331)	(27,520)
Others	(2,858)	(7,059)	(8,058)	9,264	33,600
Financing CF	(13,633)	(42,359)	(48,474)	(61,300)	(81,420)
FX & other adj	0	0	0	0	0
Chg in cash	3,891	1,309	2,016	49,831	90,770
Beginning cash	3,389	7,280	8,589	10,605	60,436
End cash	7,280	8,589	10,605	60,436	151,206
FCF	27,021	60,853	96,676	151,821	192,191

RATIO & PER SHARE DATA

FYE Jan (US$ mn)	FY24A	FY25A	FY26A	FY27E	FY28E
Gross margin (%)	72.7	75.0	71.1	74.4	72.5
EBITDA margin (%)	57.0	64.6	65.9	72.5	65.1
Op. Profit margin (%)	54.1	62.4	60.4	65.5	63.8
Net Profit margin (%)	48.8	55.8	55.6	59.2	53.0
ROE (%)	91.5	119.2	101.5	101.3	83.6
ROA (%)	55.7	82.2	75.4	77.3	68.3
Gross gearing (%)	34.6	28.9	23.9	23.5	15.3
Net gearing (%)	-35.3	-41.9	-32.8	-34.4	-34.7
Asset turn (x)	1.1	1.5	1.4	1.3	1.3
Leverage (x)	1.6	1.4	1.3	1.3	1.2
Payable days	43	50	47	57	58
Creditors days	41	46	52	53	58
Inventory days	115	86	92	128	190
EPS, diluted	1.2	2.9	4.9	10.0	16.7
BVPS	1.7	3.2	6.4	13.2	26.7
Net cash per share	0.6	1.3	2.1	4.5	9.3
SPS	2.4	5.3	8.8	16.8	31.6
FCFPS	1.1	2.6	4.5	8.0	10.0
DPS	0.02	0.03	0.04	1.00	1.20

Source: NVDA, Aletheia/TAG

Advanced Micro Devices

↔ Buy

#Semiconductor #DataCenter #AI #CPU

Price: **$ 611.76** Target Price: **$ 700.00** Upside: **14.4%** Total return: **14.4%** Risk: High

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Advanced Micro Devices

Bloomberg	AMD US
RIC	AMD.QQ
Market cap (USDbn)	992
Average daily T/O (USD|bn)	13.6

The Muse Impact

The CPU trade has come out in investors' favour again after Meta debuted its consumer-facing agent Muse. We welcome the positives, but we'd also like to remind investors that upside to our earnings estimates for AMD will likely only come in 2028E because 2027E is supply constrained. Upside to 2027E, if any, has to be on AMD's more aggressive pricing, which we haven't seen it yet. For the slighter longer term, however, upside drivers are not only CPUs but also GPUs and potentially ASICs. This should form a multi-year growth trajectory for AMD beyond 2027E. We maintain BUY for AMD and raise our TP to $700.

2027E upside is supply constrained, 2028E to be seen

We expect limited upside to our 2027E for server CPU revenue/shipments of $40b/13.5m for AMD as it is supply constrained. The following year, 2028E, should be better as we expect TSMC's N2 and ASE's FOCoS-Bridge output to grow by 50%/100%+ YoY, respectively. As such, we are modelling AMD's server CPU revenue and shipments to still grow by 70%/30% YoY in 2028E, above the 60%-70% CAGR in 2025-2030E. AMD remains as the major beneficiary of agentic AI growth, especially for consumer-facing agents, which is mostly run on x86 CPUs.

GPU upside in 2028E; potential TPUs in 2029E onwards

We expect AMD to ship 5GW+ MI450X/455X GPUs in 2027E vs our previous ~4.5GW forecast, driven by incremental demand from Meta, despite lower GMs. Looking into 2028E, we believe demand for AI GPUs will double to 10GW, up from our previous 7GW -8GW forecast, driven by an increase in demand from Anthropic and OAI. AMD's AI GPU revenue could reach $100b+ if all the demand materializes into real shipments. In addition, we believe AMD is approaching Google to bid TPU, starting from TPUv10. We believe AMD has a good value proposition and can potentially share some TPU content – if not win a whole project along with its partner MRVL – directly. The TPU contribution will only start from the outer year of our forecast horizon but is helpful for further growth and multiple.

Good momentum and positioning with rich multiple. How to trade?

AMD's growth potential is the highest among the compute names in our coverage. However, expectation is also not low, evidenced by its 31x/18x 2027/28E P/E, the highest among peers. This also makes it risky to own the shares into earnings. AMD is definitely the stock to own in a risk-on mode to ride the agentic AI momentum, but investors always need to keep its rich multiple in mind. An ease in supply tightness (more likely in 2028E) or AMD successfully raising prices further will make us more aggressive on the shares.

KEY FINANCIAL AND VALUATION MATRIX

FYE Dec (US$ mn, Non-GAAP)	FY22A	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	23,601	22,680	25,785	34,639	51,450	118,112	191,354
OPN Profit	6,345	4,854	6,138	7,768	14,734	40,054	72,407
Net Profit	5,504	4,302	5,420	6,831	13,034	35,032	63,179
Non-GAAP EPS ($), Diluted	3.54	2.65	3.32	4.17	7.86	19.71	34.62
Consensus Non-GAAP EPS ($)	-	-	-	-	7.63	15.88	22.87
PER (X) Non-GAAP	172.8	230.9	184.3	146.7	77.9	31.0	17.7
PBR (X)	18.1	17.8	17.4	16.0	13.7	10.4	6.6
EV/EBITDA (X)	155.5	119.4	108.3	92.9	56.6	25.5	14.1
ROE (%)	4.2	1.5	2.9	7.2	14.0	33.5	42.1
Total Return (%)	0.0	0.1	0.1	0.1	0.1	0.0	0.0

Source: AMD, FactSet, Aletheia/TAG

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Quantifying The Muse Impact

We have yet to have Meta's official document to talk about the hardware requirement per user, but some checks (with Muse) suggested the secure VM sandbox per user allocation is 2x virtual VPUs, 8GB DRAM and 100GB eSSD. Muse is still run on AMD's Turin CPU. But if we use AMD's upcoming Venice CPU as a reference, its Venice Dense version will be equipped with 256 CPU cores. If this set of configurations is correct, every 100m Muse users will require 780k Venice CPUs, 6.4b-Gb DRAM and 10EB eSSD. Notably, these compute resources are dedicated for any specific user and are not shared.

The numbers don't appear to be exciting compared with our/Street estimates of 2027E shipments, including ~50m CPUs, 430b-Gb total DRAM density and 700-900EB eSSD density. Every 100m Muse users only require <2% of each. However, we note Facebook users are at 3b+, IG at 2b+, and ChatGPT has also exceeded 1b. The best public available information suggests there are already 3.4m+ downloads of Muse and that it is the most popular consumer-facing app to date. We believe capacity constraint (CPU shortage) actually caps the user base growth because Muse is only available in the US and Canada at the time of writing this report.

In addition, we believe all major consumer-facing CSPs and/or Apple/Google will launch their own consumer-facing personal agent to try to gain some share. The best evidence is that AWS has already banned Muse from accessing Amazon's shopping platform on behalf of users. The future is most likely where everyone has multiple agents. Assuming that in 12-24 months from now, there will be 1b consumers having their personal agents, with everyone having two agents on average, hardware demand will become 15.6m flagship CPUs, 128b-Gb DRAM and 200EB eSSD – and this is on top of original demand already exceeding supply.

TPUs – what is AMD's value proposition?

We believe AMD's value proposition is that it can do compute tile design with much lower required profit than AVGO. It is true that AMD may not have comparable I/O SerDes technology vs AVGO or even MTK, but this should not be an issue if it can work together with MRVL. For instance, we believe MRVL has a high chance of supplying AMD's scale-up ASICs starting with MI600 AI GPUs and is likely to provide CPO solutions for the MI500 inter-rack scale-up.

The studies below show how much money Google could save if it moves away from AVGO to other lower-cost suppliers such as AMD or MTK. We analyze the cost and partners' profit for TPUv8i (by MTK) and TPUv8t (by AVGO) and perform a scenario analysis if TPUv8i/v8t is partnered with AMD. In short, AVGO's hypothetical GM, if it also prices its TPU in the same way as MTK and potentially AMD, that is, excluding HBM, would be as high as 84% given the dollar profit of $9,000. This compares with MTK's 43% GM and $2,150 dollar profit. Notably, we have assumed a significant higher "other" cost for MTK because the firm needs much more IP licensing vs AVGO. Google's cost per TPUv8t is still less than half of the cost per TPUv8i despite the real cost (excluding design partners' profit) difference between these two being <$2,000 ($7,150 for TPUv8t and $9,000 for TPUv8i).

Then we dive further into the scenario analysis if TPUv8i/TPUv8t is partnered with AMD. Let's start with TPUv8i. Even if we assume that AMD's other cost for all the required IP licensing will be as high as MTK's, Google can still save $5,350 per chip if AMD only requires a 60% GM based on "HBM excluded ASP". The 24% difference in the GM rate (vs AVGO) means more than the 25% difference in cost. Such math should encourage Google to aggressively move from AVGO as long as either its internal team or new design partner (aka AMD) can make up AVGO's design work. On the other hand, moving from MTK to AMD will cost $2,125 more per chip or a 23% increase. Unless Google thinks this extra cost is justified because it can save the internal R&D resources. switching from MTK to AMD doesn't make sense.

The real outcome may not necessarily be binary. For instance, AVGO still has plenty of room to cut prices to secure its share while still keeping its gross margin rate. Google may also have AMD share some of the design work for AVGO's future TPUs as long as Google can integrate work from both firms seamlessly. It is also possible that Google could outsource its compute tile design for MTK's TPU (which Google does internally) to AMD while still asking MTK to do the rest, though we have not heard about this combination. As long as cost-saving is a key consideration for Google, which we believe to be true and exactly why it is pushing COT, the trend favors those lower-cost partners with the knowhow that Google does not have in-house.

FIGURE 1: COST ANALYSIS FOR TPUV8I AND TPUV8T (BASED ON 2027E COST)

Product code	TPU V8t Zebrafish	TPU v8i Sunfish	TPU v8t by AMD Zebrafish	TPU v8i by AMD Sunfish	Remark
Content Analysis					
Logic tile	700	750	700	750	TSMC's wafer
CoWoS	750	950	750	950	20/16 per CoWoS for TPUv8t/TPUv8i
Others	1,400	400	1,400	1,400	MTK/AMD higher for more IP licensing
Total non-Memory cost	2,850	2,100	2,850	3,100	
HBM	4,300	6,900	4,300	6,900	$2.5/Gb, could be higher
Partner's profit	2,150	11,000	4,275	4,650	
GOOGL's cost	**9,300**	**20,000**	**11,425**	**14,650**	
Partner's ASP					
HBM included*	9,300	20,000	11,425	14,650	AVGO prices including HBM
GM	*23%*	***55%***	*37%*	*32%*	
HBM excluded^	**5,000**	13,100	**7,125**	**7,750**	MTK (and AMD) prices excluding HBM
GM	***43%***	*84%*	***60%***	***60%***	
Cost vs existing vendor			*23%*	*-27%*	23% more vs MTK but 27% less vs AVGO

Source: Aletheia Capital/TAG *if ASP by MTK (and AMD) also includes HBM ^if AVGO's ASP exclude HBM

Maintain BUY; Raise TP to $700

Earnings revision: upside mainly in 2028E

We tweak our 2026/27E EPS modestly to factor in the strong CPU demand and supply constraint in the next 15 months. At the same time, we are revising up our 2028E earnings aggressively by 27% and forecast AMD's 2028E EPS can reach ~$35. The key driver for our revision is AI GPUs, of which we now assume AMD can ship up to 10GW in 2028E vs ~5GW in 2027E. We have previously already assumed AMD's CPU revenue/units can grow by 70%/30% YoY and we are keeping our estimates unchanged. Our 2027/2028E EPS forecast is 24%/52% ahead of the Factset consensus but should be in the range of the current bullish buyside expectations.

Maintain BUY, Raise TP to $700

We have previously adopted an SOTP valuation for AMD given the different growth trajectory and margins profile among its different business segments. Should our forecast for 2027/2028E prove to be correct, AMD's data center business will account for 80%+ of its 2027/28E earnings from 50%/65% in 2025A/2026E. As such, we are moving our SOTP to straight P/E which is simpler and more straightforward.

Our TP is based on 26x average 2027/28E P/E and implies a 20x P/E on a 2028E basis. This is about the same as our target multiple for MediaTek (2454 TT, BUY, TP: TWD6,500) because both of them are showing the strongest earnings growth in our forecast horizon, despite different drivers behind each. To raise our TP further for a more constructive view for AMD, we need to roll-over to a 2028E valuation base with the same multiple – which we believe is premature because the growth trajectory for outer years remains unclear.

FIGURE 2: ALETHEIA'S EARNINGS REVISION FOR AMD

	3Q26E			4Q26E			FY26E			FY27E			FY28E		
US$ bn	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)
Sales	13.0	12.1	7.5	16.7	17.0	(1.8)	51.5	50.6	1.7	118.1	111.2	6.2	191.4	163.5	17.0
Gross profit	7.3	6.9	7.2	9.3	9.4	(1.3)	28.8	28.3	1.9	59.8	58.2	2.8	100.4	85.0	18.1
Operating profit	3.7	3.5	7.3	5.4	5.6	(4.4)	14.7	14.7	0.5	40.1	38.1	5.1	72.4	56.9	27.2
Net profit	3.3	3.0	8.9	4.7	4.9	(3.4)	13.0	12.9	1.3	35.0	33.2	5.7	63.2	49.5	27.6
EPS (US$)	1.97	1.83	7.8	2.86	2.98	(4.2)	7.86	7.81	0.6	19.71	19.46	1.3	34.62	27.37	26.5
Gross margins (%)	56.5	56.7	(0.2)	55.9	55.6	0.3	56.0	55.9	0.1	50.6	52.3	(1.7)	52.5	52.0	0.5
OP margins (%)	28.5	28.6	(0.1)	32.4	33.2	(0.9)	28.6	29.0	(0.3)	33.9	34.3	(0.3)	37.8	34.8	3.0
Net margins (%)	25.1	24.8	0.3	28.4	28.9	(0.5)	25.3	25.4	(0.1)	29.7	29.8	(0.1)	33.0	30.3	2.7

Source: AMD, Aletheia Capital/TAG

FIGURE 3: ALETHEIA'S EARNINGS ESTIMATES FOR AMD VS CONSENSUS

	3Q26E			4Q26E			FY26E			FY27E			FY28E		
US$ bn	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)
Sales	13.0	13.0	0.2	16.7	16.2	3.0	51.5	50.9	1.0	118.1	89.4	32.2	191.4	123.1	55.5
Gross profit	7.3	7.3	0.8	9.3	9.0	3.9	28.8	28.2	2.0	59.8	49.3	21.4	100.4	67.9	48.0
Operating profit	3.7	3.6	2.2	5.4	5.1	6.6	14.7	14.4	2.5	40.1	30.2	32.7	72.4	43.8	65.4
Net profit	3.3	3.2	2.1	4.7	4.4	7.4	13.0	12.5	4.1	35.0	26.5	32.0	63.2	38.6	63.7
EPS (US$)	1.97	1.93	2.2	2.86	2.68	6.6	7.86	7.63	3.0	19.71	15.88	24.1	34.62	22.87	51.4
Gross margins (%)	56.5	56.1	0.4	55.9	55.4	0.5	56.0	55.5	0.6	50.6	55.2	(4.5)	52.5	55.2	(2.7)
OP margins (%)	28.5	27.9	0.6	32.4	31.3	1.1	28.6	28.2	0.4	33.9	33.8	0.1	37.8	35.6	2.3
Net margins (%)	25.1	24.7	0.5	28.4	27.3	1.2	25.3	24.6	0.7	29.7	29.7	(0.0)	33.0	31.4	1.7

Source: FactSet, AMD, Aletheia Capital /TAG

Earnings model

Revenue ($, mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Server	2,174	2,388	2,641	2,585	3,265	4,014	4,835	5,550	7,750	8,000	10,700	13,000	9,788	17,664	39,450	67,326
AI (UBB+Racks)	1,500	852	1,700	2,795	2,510	2,630	2,900	6,525	10,000	11,250	17,500	17,450	6,847	14,565	56,200	98,250
Xilinx	823	824	857	950	873	977	1,221	1,160	1,160	1,160	1,218	1,279	3,454	4,231	4,818	5,311
Clients	2,294	2,499	2,750	3,097	2,885	3,062	3,188	2,739	2,644	2,866	3,227	3,249	10,640	11,874	11,986	13,245
Gaming and others	647	1,122	1,298	843	720	853	863	679	685	1,306	1,923	1,743	3,910	3,116	5,658	7,221
Total revenue	7,438	7,685	9,246	10,270	10,253	11,536	13,008	16,654	22,239	24,583	34,569	36,721	34,639	51,450	118,112	191,354
Revenue mix																
Server	29%	31%	29%	25%	32%	35%	37%	33%	35%	33%	31%	35%	28%	34%	33%	35%
AI (UBB+Racks)	20%	11%	18%	27%	24%	23%	22%	39%	45%	46%	51%	48%	20%	28%	48%	51%
Xilinx	11%	11%	9%	9%	9%	8%	9%	7%	5%	5%	4%	3%	10%	8%	4%	3%
Clients	31%	33%	30%	30%	28%	27%	25%	16%	12%	12%	9%	9%	31%	23%	10%	7%
Gaming and others	9%	15%	14%	8%	7%	7%	7%	4%	3%	5%	6%	5%	11%	6%	5%	4%
Total revenue	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

P&L ($mn, FYE Dec, Non-GAAP)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	7,438	7,685	9,246	10,270	10,253	11,536	13,008	16,654	22,239	24,583	34,569	36,721	34,639	51,450	118,112	191,354
COGS	-3,446	-4,359	-4,254	-4,415	-4,568	-5,048	-5,664	-7,350	-10,730	-12,106	-17,420	-18,037	-16,474	-22,630	-58,293	-90,905
Gross profit	3,992	3,326	4,992	5,855	5,685	6,488	7,343	9,304	11,510	12,476	17,149	18,684	18,165	28,820	59,819	100,450
OPEX	-2,213	-2,429	-2,754	-3,001	-3,145	-3,394	-3,636	-3,911	-4,424	-4,747	-5,102	-5,492	-10,397	-14,086	-19,765	-28,043
OP profit	1,779	897	2,238	2,854	2,540	3,094	3,707	5,393	7,086	7,730	12,047	13,191	7,768	14,734	40,054	72,407
Non-OP profit	19	60	45	322	128	561	53	53	53	53	53	53	446	795	212	212
PTI	1,798	957	2,283	3,176	2,668	3,655	3,760	5,446	7,139	7,783	12,100	13,244	8,214	15,529	40,266	72,619
Tax	-232	-176	-318	-657	-403	-895	-489	-708	-928	-1,012	-1,573	-1,722	-1,383	-2,495	-5,235	-9,440
Net profit	1,566	781	1,965	2,519	2,265	2,760	3,271	4,738	6,211	6,771	10,527	11,523	6,831	13,034	35,032	63,179
FD Non-GAAP EPS, $	**0.96**	**0.48**	**1.20**	**1.53**	**1.37**	**1.66**	**1.97**	**2.86**	**3.74**	**3.89**	**5.77**	**6.31**	**4.17**	**7.86**	**19.71**	**34.62**
FD, GAAP EPS, $	**0.44**	**0.54**	**0.75**	**0.92**	**0.84**	**1.38**	**1.34**	**2.23**	**3.03**	**3.21**	**5.13**	**5.68**	**2.65**	**5.80**	**17.05**	**31.81**

Profitability	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Gross margin	53.7%	43.3%	54.0%	57.0%	55.4%	56.2%	56.5%	55.9%	51.8%	50.8%	49.6%	50.9%	52.4%	56.0%	50.6%	52.5%
OPN margin	23.9%	11.7%	24.2%	27.8%	24.8%	26.8%	28.5%	32.4%	31.9%	31.4%	34.8%	35.9%	22.4%	28.6%	33.9%	37.8%
PTI margin	24.2%	12.5%	24.7%	30.9%	26.0%	31.7%	28.9%	32.7%	32.1%	31.7%	35.0%	36.1%	23.7%	30.2%	34.1%	38.0%
Net margin	21.1%	10.2%	21.3%	24.5%	22.1%	23.9%	25.1%	28.4%	27.9%	27.5%	30.5%	31.4%	19.7%	25.3%	29.7%	33.0%

QoQ chg.	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	-3%	3%	20%	11%	0%	13%	13%	28%	34%	11%	41%	6%	-	-	-	-
Gross profit	-4%	-17%	50%	17%	-3%	14%	13%	27%	24%	8%	37%	9%	-	-	-	-
OPN profit	-12%	-50%	149%	28%	-11%	22%	20%	45%	31%	9%	56%	9%	-	-	-	-
PTI	-12%	-47%	139%	39%	-16%	37%	3%	45%	31%	9%	55%	9%	-	-	-	-
Net profit	-12%	-50%	152%	28%	-10%	22%	19%	45%	31%	9%	55%	9%	-	-	-	-

YoY chg.	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	36%	32%	36%	34%	38%	50%	41%	62%	117%	113%	166%	120%	34%	49%	130%	62%
Gross profit	40%	7%	37%	41%	42%	95%	47%	59%	102%	92%	134%	101%	32%	59%	108%	68%
OPN profit	57%	-29%	30%	41%	43%	245%	66%	89%	179%	150%	225%	145%	27%	90%	172%	81%
PTI	55%	-26%	32%	55%	48%	282%	65%	71%	168%	113%	222%	143%	32%	89%	159%	80%
Net profit	55%	-31%	31%	42%	45%	253%	66%	88%	174%	145%	222%	143%	26%	91%	169%	80%

Source: Company, Aletheia Capital/TAG

Financial Statements

INCOME STATEMENT

FYE Dec ($ mn), Non-GAAP	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	25,785	34,639	51,450	118,112	191,354
COGS	-12,026	-16,474	-22,630	-58,293	-90,905
Gross profit	13,759	18,165	28,820	59,819	100,450
Operating expense	-7,621	-10,397	-14,086	-19,765	-28,043
EBITDA	9,202	10,772	17,765	43,076	75,432
Operating profit	6,138	7,768	14,734	40,054	72,407
Interest income (expense)	-92	-131	-148	-148	-148
Other income (expense)	181	577	943	360	360
Profit before tax	6,227	8,214	15,529	40,266	72,619
Tax	-807	-1,383	-2,495	-5,235	-9,440
Net profit	5,420	6,831	13,034	35,032	63,179
GAAP EPS $, Diluted	0.99	2.65	5.80	17.05	31.81
Non-GAAP EPS $, Diluted	3.32	4.17	7.86	19.71	34.62
Change YoY					
Revenue	14%	34%	49%	130%	62%
Gross profit	20%	32%	59%	108%	68%
Operating profit	26%	27%	90%	172%	81%
Net profit	26%	26%	91%	169%	80%
EPS (non-GAAP)	25%	26%	88%	151%	76%

BALANCE SHEET

FYE Dec ($ mn)	FY24A	FY25A	FY26E	FY27E	FY28E
Cash/ST investment	5,132	10,552	11,859	21,112	53,704
Inventories	5,734	7,920	12,150	29,491	48,679
Receivables	6,192	6,315	12,123	26,897	45,629
Current assets	19,049	26,947	38,795	80,162	150,674
Fixed assets	1,802	2,312	3,651	4,208	4,777
Goodwill & intangibles	43,769	41,831	40,034	37,956	35,959
Other LT assets	4,606	5,836	8,398	8,398	8,398
LT assets	50,177	49,979	52,083	50,562	49,134
Total assets	69,226	76,926	90,878	130,725	199,808
Payable	1,990	2,929	4,835	11,727	19,629
Other liabilities	5,291	6,526	6,722	6,722	6,722
Current liabilities	7,281	9,455	11,557	18,449	26,351
LT debt	1,721	2,348	2,351	2,351	2,351
other LT Liabilities	2,656	2,124	2,808	2,808	2,808
LT liabilities	4,377	4,472	5,159	5,159	5,159
Total liabilities	11,658	13,927	16,716	23,608	31,510
Equity	57,568	62,999	74,162	107,116	168,298
Liability&Equity	69,226	76,926	90,878	130,725	199,808

CASH FLOW

FYE Dec ($ mn)	FY24A	FY25A	FY26E	FY27E	FY28E
Net profit	1,641	4,335	9,612	30,339	58,043
Dep. & amortization	3,064	3,004	3,031	3,022	3,024
Chg in WC	-3,432	-1,900	-8,393	-25,222	-30,018
Others	1,768	2,270	469	2,616	3,139
OPN cash flow	3,041	7,709	4,720	10,754	34,188
Capex	-636	-974	-1,847	-1,501	-1,596
Chg in investment					
Others	-465	-4,559	-4,218	0	0
Investing CF	-1,101	-5,533	-6,065	-1,501	-1,596
Chg in debt	-750	-950	0	0	0
Equity raised (Repurchased)	-862	-1,316	-562	0	0
Dividend paid					
Others	-450	1,835	197	0	0
Financing CF	-2,062	-431	-365	0	0
FX & other adj					
Chg in cash	-122	1,745	-1,711	9,253	32,592
Beginning cash	16,133	19,148	19,343	16,936	106,734
End cash	16,011	20,893	17,633	26,189	139,325
FCF	2,405	6,735	2,872	9,253	32,592

RATIO & PER SHARE DATA

FYE Dec ($)	FY24A	FY25A	FY26E	FY27E	FY28E
Gross margin (%)	53.4	52.4	56.0	50.6	52.5
EBITDA margin (%)	35.7	31.1	34.5	36.5	39.4
Op. Profit margin (%)	238.0	224.3	286.4	339.1	378.4
Net Profit margin (%)	21.0	19.7	25.3	29.7	33.0
ROE (%)	2.9	7.2	14.0	33.5	42.1
ROA (%)	2.4	5.9	11.5	27.4	35.1
Gross gearing (%)	16.8	18.1	18.4	18.1	15.8
Net gearing (%)	-5.9	-13.0	-12.8	-17.5	-30.5
Asset turnover (x)	0.4	0.5	0.6	0.9	1.0
Leverage (x)	1.2	1.2	1.2	1.2	1.2
Payable days	62	54	63	52	63
Receivable days	82	66	65	60	69
Inventory days	153	151	162	130	157
EPS, diluted	3.3	4.2	7.9	19.7	34.6
BVPS	35	38	45	59	92
Net cash per share	2	5	6	10	28
SPS	16	21	31	65	105
FCFPS	1	4	2	5	18
Total returns (%)	0	0	0	0	0

Source: Company, Aletheia Capital /TAG

Broadcom

#AI #ASIC

↔ Buy

Price: **$ 351.19** Target Price: **$ 500** Upside: **42%** Total return: **47%** Risk: High

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Broadcom

Bloomberg	AVGO US
RIC	AVGO.QQ
Market cap (USDbn)	1,669
Average daily T/O (USD|bn)	10.0

Share Loss and Diversification

AVGO's shares have done nothing despite the Street upgrading 2027E earnings by 40%+ YTD, significantly below +70% for SOX. The P/E contraction is easy to understand because its share loss in TPU is materializing. We are also aware that the Street is discounting its non-TPU ASIC businesses, particularly OAI in 2028E, which should make for clearer visibility in the next two quarters. If everything goes smoothly, TPU could only account for half of its ASIC business in 2028E from 80%-90% now, and its $30 FY28E EPS guidance is achievable. AVGO is not our preferred pick for compute, but we think its diversification and earnings growth are under-appreciated. We maintain BUY with a tweaked TP of $500.

10GW/20GW in 2028E: OAI is the key

We believe 10GW in 2027E guidance is less of a concern. Its TPU alone, including Anthropic and Google's demand, could already reach 7-8GW in 2027E. With OAI's Jalapeno contributing another 1-1.5GW and 1GW from Meta, we are already about at the guidance. 20GW in 2028E, from the supply angle, is not impossible as TSMC should have enough CoWoS and AVGO has requested more than enough HBM. We just need to continue to follow up if there are corresponding projects in the supply chain. We believe 10GW for Anthropic and ~2GW is less of a concern, but 5GW for OAI is critical. In the current supply tightness environment, AVGO and OAI may need to confirm with suppliers in 1Q27E at the latest to secure 2028E supply.

TPU shares loss is inevitable

It is already well-known that AVGO will lose its sole vendor position for Google's TPU starting in 4Q26E, the only question is how fast market share will shift. We expect AVGO's 2027E TPUv8 shipments will be on a par with MTK's despite its ~80% revenue share, but MTK could overtake AVGO in both shipments and compute power in 2028E. More critically, there is a high probability that AMD will share some of its value in TPUv10. A detailed workload and value breakdown remains to be finalized, and the impact will only happen in 2029E, which is beyond our forecast horizon. But AVGO's P/E multiple should be adversely impacted.

Maintain BUY, lower TP to $500 on lowered multiple

We maintain BUY for AVGO but lower our TP to $500 (20x avg FY27/28E P/E) from $525 (25x FY27E P/E). Stocks should gradually price in 2028E because the outlook for 2028E earnings and market share is getting clearer. We lower our target P/E to 20x to factor-in concern about share loss, the growth rate slowing down (though still at an astonishing 37% YoY for FY28E), and a market cap penalty (just like NVDA). Admittedly, we are also seeking the right multiple for AVGO given its set-up, but a discount to 16x/12x FY27/28E P/E looks too demanding to us.

KEY FINANCIAL AND VALUATION MATRIX

FYE Oct (US$ mn, Non-GAAP)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	35,819	51,574	63,887	106,168	199,924	276,491
AI revenue	*3,885*	*12,200*	*20,200*	*57,697*	*144,193*	*215,234*
AI revenue as of total semi	*14%*	*41%*	*55%*	*77%*	*89%*	*91%*
OPN Profit	22,125	30,736	41,997	71,248	128,585	174,974
Net Profit	18,379	23,733	33,728	57,997	106,849	146,001
Non-GAAP EPS ($), Diluted	4.23	4.86	6.82	11.73	21.63	29.55
Consensus Non-GAAP EPS ($)	-	-	-	11.67	19.25	30.53
PER (X) Non-GAAP	83.1	72.2	51.5	29.9	16.2	11.9
PBR (X)	62.6	24.6	21.0	15.1	9.9	6.8
EV/EBITDA (X)	58.8	42.3	34.6	21.7	12.2	8.7
ROE (%)	77%	52%	45%	60%	75%	69%

Source: company data, FactSet, Aletheia/TAG

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TPUs – a Likely Shift in TPUv10

We believe AMD's value proposition is that it can do compute tile design with much lower required margins than AVGO. It is true that AMD may not have comparable I/O SerDes technology vs AVGO or even MTK, but this should not be an issue if it can work together with MRVL. For instance, we believe MRVL has a high chance of supplying AMD's scale-up ASICs starting with MI600 AI GPUs and is likely to provide CPO solutions for the MI500 inter-rack scale-up.

The studies below show how much money Google could save if it moves away from AVGO to other lower-cost suppliers such as AMD or MTK. We analyze the cost and partners' profit for TPUv8i (by MTK) and TPUv8t (by AVGO) and perform a scenario analysis if TPUv8i/v8t is partnered with AMD. In short, AVGO's hypothetical GM, if it also prices its TPU in the same way as MTK and potentially AMD, that is, excluding HBM, would be as high as 84% given the dollar profit of $9,000. This compares with MTK's 43% GM and $2,150 dollar profit. Notably, we have assumed a significant higher "other" cost for MTK because the firm needs much more IP licensing vs AVGO. Google's cost per TPUv8t is still less than half the cost per TPUv8i despite the real cost (excluding design partners' profit) difference between these two being <$2,000 ($7,150 for TPUv8t and $9,000 for TPUv8i).

Then we dive further into the scenario analysis if TPUv8i/TPUv8t is partnered with AMD. Let's start with TPUv8i. Even if we assume that AMD's other cost for all the required IP licensing will be as high as MTK's, Google can still save $5,350 per chip if AMD only requires a 60% GM based on "HBM excluded ASP". The 24% difference in the GM rate (vs AVGO) means more than the 25% difference in cost. Such math should encourage Google to aggressively move from AVGO as long as either its internal team or new design partner (aka AMD) can make up AVGO's design work. On the other hand, moving from MTK to AMD will cost $2,125 more per chip or a 23% increase. Unless Google thinks this extra cost is justified because it can save the internal R&D resources. switching from MTK to AMD doesn't make sense.

The real outcome may not necessarily be binary. For instance, AVGO still has plenty of room to cut prices to secure its share while still keeping its gross margin rate. Google may also have AMD share some of the design work for AVGO's future TPUs as long as Google can integrate work from both firms seamlessly. It is also possible that Google could outsource its compute tile design for MTK's TPU (which Google does internally) to AMD while still asking MTK to do the rest, though we have not heard about this combination. As long as cost-saving is a key consideration for Google, which we believe to be true and exactly why it is pushing COT, the trend favors those lower-cost partners with the knowhow that Google does not have in-house.

FIGURE 1: COST ANALYSIS FOR TPUV8I AND TPUV8T (BASED ON 2027E COST)

Product code	TPU V8t Zebrafish	TPU v8i Sunfish	TPU v8t by AMD Zebrafish	TPU v8i by AMD Sunfish	Remark
Content Analysis					
Logic tile	700	750	700	750	TSMC's wafer
CoWoS	750	950	750	950	20/16 per CoWoS for TPU v8t/TPU v8i
Others	1,400	400	1,400	1,400	MTK/AMD higher for more IP licensing
Total non-Memory cost	2,850	2,100	2,850	3,100	
HBM	4,300	6,900	4,300	6,900	$2.5/Gb, could be higher
Partner's profit	2,150	11,000	4,275	4,650	
GOOGL's cost	**9,300**	**20,000**	**11,425**	**14,650**	
Partner's ASP					
HBM included*	9,300	20,000	11,425	14,650	AVGO prices including HBM
GM	*23%*	***55%***	*37%*	*32%*	
HBM excluded^	**5,000**	13,100	**7,125**	**7,750**	MTK (and AMD) prices excluding HBM
GM	***43%***	*84%*	***60%***	***60%***	
Cost vs existing vendor			*23%*	*-27%*	23% more vs MTK but 27% less vs AVGO

Source: Aletheia Capital/TAG *if ASP by MTK (and AMD) also includes HBM ^if AVGO's ASP exclude HBM

FIGURE 2: BROADCOM - ESTIMATE REVISION

	4Q26E			1Q27E			FY26E			FY27E		
(US$ bn)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)	New	Prev	Var (%)
Sales	35.1	30.7	14.4	35.8	35.0	2.3	106.2	97.8	8.5	199.9	195.1	2.5
Gross profit	25.6	22.7	12.8	26.3	25.5	2.9	79.8	74.1	7.7	138.3	134.7	2.7
OP profit	23.4	20.5	14.4	24.0	23.2	3.3	71.2	65.5	8.8	128.6	125.0	2.8
Net profit	19.4	16.9	14.5	19.8	19.3	3.0	58.0	53.5	8.4	106.8	104.4	2.3
EPS (US$)	3.92	3.41	15.0	4.02	3.89	3.3	11.73	10.79	8.7	21.63	21.06	2.7
Gross margin (%)	73.1	74.1	-1.0	73.3	72.9	0.4	75.2	75.8	-0.6	69.2	69.0	0.1
OP margin (%)	66.7	66.7	0.0	67.0	66.4	0.6	67.1	67.0	0.1	64.3	64.1	0.2
Net margin (%)	55.2	55.1	0.1	55.4	55.0	0.4	54.6	54.7	0.0	53.4	53.5	-0.1

Source: Broadcom, Aletheia/TAG

FIGURE 3: BROADCOM - ESTIMATE VS CONSENSUS

	4Q26E			1Q27E			FY26E			FY27E		
(US$ bn)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)
Sales	35.1	34.9	0.6	35.8	37.1	-3.4	106.2	106.0	0.2	199.9	173.7	15.1
Gross profit	25.6	25.4	1.0	26.3	27.1	-3.1	79.8	79.6	0.3	138.3	124.4	11.2
OP profit	23.4	23.1	1.1	24.0	24.5	-1.8	71.2	70.9	0.5	128.6	113.8	13.0
Net profit	19.4	18.9	2.7	19.8	20.1	-1.2	58.0	57.7	0.6	106.8	94.3	13.3
EPS (US$)	3.92	3.83	2.4	4.02	4.10	-2.0	11.73	11.67	0.5	21.63	19.25	12.4
Gross margin (%)	73.1	72.8	0.3	73.3	73.0	0.3	75.2	75.1	0.1	69.2	71.6	-2.4
OP margin (%)	66.7	66.4	0.3	67.0	65.9	1.1	67.1	66.9	0.2	64.3	65.5	-1.2
Net margin (%)	55.2	54.1	1.1	55.4	54.2	1.2	54.6	54.4	0.2	53.4	54.3	-0.8

Source: Broadcom, Aletheia/TAG

Earnings Model

Revenue model	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue breakdown ($m)																
Semiconductor	8,212	8,408	9,166	11,072	12,515	15,009	20,839	26,327	26,413	41,710	41,876	52,723	36,858	74,690	162,721	237,312
AI	4,100	4,400	5,200	6,500	8,440	10,800	16,700	21,757	21,899	37,368	37,368	47,559	20,200	57,697	144,193	215,234
QoQ	*16%*	*7%*	*18%*	*25%*	*30%*	*28%*	*55%*	*30%*	*1%*	*71%*	*0%*	*27%*				
YoY	*78%*	*42%*	*60%*	*83%*	*106%*	*145%*	*221%*	*235%*	*159%*	*246%*	*124%*	*119%*	*66%*	*186%*	*150%*	*49%*
Custom ASICs	2,460	2,640	3,380	4,550	5,655	6,480	12,191	13,054	13,140	25,410	25,410	32,340	13,030	37,380	96,300	146,359
Networking	1,640	1,760	1,820	1,950	2,785	4,320	4,509	8,703	8,760	11,958	11,958	15,219	7,170	20,317	47,894	68,875
Non-AI	4,112	4,008	3,966	4,572	4,075	4,209	4,139	4,570	4,513	4,342	4,508	5,164	16,658	16,993	18,527	22,078
Infrastructure software	6,704	6,596	6,786	6,943	6,796	7,178	8,752	8,752	9,411	9,411	9,191	9,191	27,029	31,478	37,204	39,179
Total revenue	**14,916**	**15,004**	**15,952**	**18,015**	**19,311**	**22,187**	**29,591**	**35,079**	**35,823**	**51,120**	**51,067**	**61,914**	**63,887**	**106,168**	**199,924**	**276,491**
Revenue breakdown (%)																
Semiconductor	55%	56%	57%	61%	65%	68%	70%	75%	74%	82%	82%	85%	58%	70%	81%	86%
AI	27%	29%	33%	36%	44%	49%	56%	62%	61%	73%	73%	77%	32%	54%	72%	78%
Custom ASICs	16%	18%	21%	25%	29%	29%	41%	37%	37%	50%	50%	52%	20%	35%	48%	53%
Networking	11%	12%	11%	11%	14%	19%	15%	25%	24%	23%	23%	25%	11%	19%	24%	25%
Non-AI	28%	27%	25%	25%	21%	19%	14%	13%	13%	8%	9%	8%	26%	16%	9%	8%
Infrastructure software	45%	44%	43%	39%	35%	32%	30%	25%	26%	18%	18%	15%	42%	30%	19%	14%
Total revenue	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**
AI as of total semi	***50%***	***52%***	***57%***	***59%***	***67%***	***72%***	***80%***	***83%***	***83%***	***90%***	***89%***	***90%***	***55%***	***77%***	***89%***	***91%***
Segment profit																
Semiconductor																
Revenue	8,212	8,408	9,166	11,072	12,515	15,009	20,839	26,327	26,413	41,710	41,876	52,723	36,858	74,690	162,721	237,312
Gross profit	5,584	5,802	6,141	7,529	8,510	10,206	13,962	17,330	17,313	26,200	26,325	33,074	25,056	50,008	102,912	148,712
GM	*68%*	*69%*	*67%*	*68.0%*	*68.0%*	*68%*	*67%*	*66%*	*66%*	*63%*	*63%*	*63%*	*68%*	*67%*	*63%*	*63%*
Operating profit	4,706	4,806	5,217	6,503	7,503	9,100	12,770	16,125	16,129	24,901	24,924	31,658	21,232	45,498	97,612	142,484
OPM	*57%*	*57%*	*57%*	*59%*	*60%*	*61%*	*61%*	*61%*	*61%*	*60%*	*60%*	*60%*	*58%*	*61%*	*60%*	*60%*
Infrastructure software																
Revenue	6,704	6,596	6,786	6,943	6,796	7,178	8,752	8,752	9,411	9,411	9,191	9,191	27,029	31,478	37,204	39,179
Gross profit	6,212	6,109	6,358	6,510	6,358	6,903	8,229	8,314	8,940	8,940	8,732	8,732	25,189	29,804	35,343	37,220
GM	*93%*	*93%*	*94%*	*94%*	*94%*	*96%*	*94%*	*95%*	*95%*	*95%*	*95%*	*95%*	*93%*	*95%*	*95%*	*95%*
Operating profit	5,122	4,987	5,238	5,418	5,323	5,828	7,325	7,275	7,880	7,859	7,628	7,606	20,765	25,751	30,973	32,490
OPM	*76%*	*76%*	*77%*	*78%*	*78%*	*81%*	*84%*	*83%*	*84%*	*84%*	*83%*	*83%*	*77%*	*82%*	*83%*	*83%*

Non-GAAP P&L ($m)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Sales	14,916	15,004	15,952	18,015	19,311	22,187	29,591	35,079	35,823	51,120	51,067	61,914	63,887	106,168	199,924	276,491
CoGS	-3,120	-3,093	-3,453	-3,976	-4,443	-5,078	-7,400	-9,434	-9,570	-15,980	-16,010	-20,108	-13,642	-26,355	-61,669	-90,559
Gross Profit	11,796	11,911	12,499	14,039	14,868	17,109	22,191	25,644	26,253	35,140	35,056	41,806	50,245	79,812	138,255	185,932
OPEX	-1,968	-2,118	-2,044	-2,118	-2,042	-2,181	-2,096	-2,245	-2,244	-2,381	-2,504	-2,542	-8,248	-8,564	-9,670	-10,958
OP Profit	9,828	9,793	10,455	11,921	12,826	14,928	20,095	23,399	24,009	32,759	32,552	39,264	41,997	71,248	128,585	174,974
Total Non-OP Profit	-732	-738	-736	-625	-628	-627	-605	-703	-703	-703	-703	-703	-2,831	-2,563	-2,812	-2,812
Pretex Profit	9,096	9,055	9,719	11,296	12,198	14,301	19,490	22,696	23,306	32,056	31,849	38,561	39,166	68,685	125,773	172,162
Tax expense	-1,273	-1,268	-1,368	-1,582	-2,013	-2,227	-3,118	-3,331	-3,462	-4,796	-4,803	-5,864	-5,491	-10,689	-18,924	-26,161
Net Profit	7,823	7,787	8,404	9,714	10,185	12,074	16,372	19,366	19,845	27,260	27,046	32,697	33,728	57,997	106,849	146,001
Non-GAAP EPS	**1.60**	**1.58**	**1.69**	**1.95**	**2.05**	**2.44**	**3.31**	**3.92**	**4.02**	**5.52**	**5.47**	**6.62**	**6.82**	**11.73**	**21.63**	**29.55**
GAAP EPS	**1.14**	**1.03**	**0.85**	**1.74**	**1.50**	**1.91**	**2.68**	**3.51**	**3.44**	**4.95**	**4.89**	**6.11**	**4.76**	**9.60**	**19.40**	**27.42**
Profit Margins (%)																
Gross Margins	79%	79%	78%	78%	77%	77%	75%	73%	73%	69%	69%	68%	79%	75%	69%	67%
OP Margins	66%	65%	66%	66%	66%	67%	68%	67%	67%	64%	64%	63%	66%	67%	64%	63%
Net Margins	52%	52%	53%	54%	53%	54%	55%	55%	55%	53%	53%	53%	53%	55%	53%	53%
QoQ (%)																
Sales	6%	1%	6%	13%	7%	15%	33%	19%	2%	43%	0%	21%	-	-	-	-
Gross Profit	9%	1%	5%	12%	6%	15%	30%	16%	2%	34%	0%	19%	-	-	-	-
OP Profit	12%	0%	7%	14%	8%	16%	35%	16%	3%	36%	-1%	21%	-	-	-	-
Net Profit	12%	0%	8%	16%	5%	19%	36%	18%	2%	37%	-1%	21%	-	-	-	-
Non-GAAP EPS	13%	-1%	7%	16%	5%	19%	36%	18%	2%	37%	-1%	21%	-	-	-	-
GAAP EPS	27%	-10%	-17%	105%	-14%	27%	40%	31%	-2%	44%	-1%	25%	-	-	-	-
YoY (%)																
Sales	25%	20%	22%	28%	29%	48%	86%	95%	86%	130%	73%	76%	24%	66%	88%	38%
Gross Profit	31%	25%	24%	30%	26%	44%	78%	83%	77%	105%	58%	63%	27%	59%	73%	34%
OP Profit	44%	37%	32%	35%	31%	52%	92%	96%	87%	119%	62%	68%	37%	70%	80%	36%
Net Profit	49%	44%	37%	39%	30%	55%	95%	99%	95%	126%	65%	69%	42%	72%	84%	37%
Non-GAAP EPS	45%	44%	36%	38%	29%	55%	96%	101%	96%	126%	65%	69%	40%	72%	84%	37%
GAAP EPS	301%	133%	-312%	95%	32%	86%	214%	101%	129%	159%	83%	74%	291%	102%	102%	41%

Source: Broadcom, Aletheia/TAG

Financial Statements

INCOME STATEMENT

FYE Oct ($ bn)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	35.8	51.6	63.9	106.2	199.9	276.5
COGS	(9.1)	(12.1)	(13.6)	(26.4)	(61.7)	(90.6)
Gross profit	26.8	39.5	50.2	79.8	138.3	185.9
Operating expense	(4.6)	(8.7)	(8.2)	(8.6)	(9.7)	(11.0)
EBITDA	26.0	40.7	50.8	80.0	137.4	183.8
Operating profit	22.1	30.7	42.0	71.2	128.6	175.0
Interest income (expense)	(1.6)	(4.0)	(3.2)	(2.9)	(2.8)	(2.8)
Other income (expense)	0.5	0.6	0.4	0.3	0.0	(0.0)
Profit before tax	21.0	27.4	39.2	68.7	125.8	172.2
Tax	(2.6)	(3.6)	(5.5)	(10.7)	(18.9)	(26.2)
Net profit	18.4	23.7	33.7	58.0	106.8	146.0
GAAP EPS $, Diluted	3.29	1.24	4.77	9.60	19.40	27.42
Non-GAAP EPS $, Diluted	4.23	4.86	6.82	11.73	21.63	29.55
Change YoY						
Revenue	-	44%	24%	66%	88%	38%
Gross profit	-	47%	27%	59%	73%	34%
Operating profit	-	39%	37%	70%	80%	36%
Net profit	-	29%	42%	72%	84%	37%
EPS (non-GAAP)	-	15%	40%	72%	84%	37%

BALANCE SHEET

FYE Oct ($ bn)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Cash/ST investment	14.2	9.3	16.2	36.2	83.0	153.1
Inventory	1.9	1.8	2.3	5.7	12.2	14.8
Receivable	3.2	4.4	7.1	16.1	28.4	33.9
Current assets	20.8	19.6	31.6	67.9	133.5	211.8
Fixed assets	2.6	3.8	3.8	3.7	5.8	7.9
Goodwill & intangibles	47.5	138.5	130.1	121.9	113.1	104.2
Other LT assets	1.9	3.7	5.6	8.7	8.7	8.7
LT assets	52.0	146.1	139.5	134.3	127.6	120.8
Total assets	72.9	165.6	171.1	202.2	261.0	332.6
Payable	72.9	165.6	171.1	202.2	261.0	332.6
Deferred revenue	2.5	9.4	9.5	9.5	9.5	9.5
ST debt	1.7	1.5	3.3	2.3	2.3	2.3
Current liabilities	7.4	16.7	18.5	23.0	34.5	39.2
LT deferred revenue	0.0	5.1	3.5	3.6	3.6	3.6
LT debt	38.0	67.4	63.2	55.2	43.2	31.2
Total liabilities	48.9	98.0	89.8	88.6	88.1	80.8
Retained earnings	2.7	0.0	9.8	36.1	111.4	206.3
Equity	24.0	67.7	81.3	113.7	172.9	251.8
Liability&Equity	72.9	165.6	171.1	202.2	261.0	332.6

CASH FLOW

FYE Oct ($ bn)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Net profit	14.1	5.9	23.1	46.9	94.8	134.0
Dep. & amortization	3.8	10.0	8.8	8.7	8.9	8.9
Chg in WC	(0.9)	(2.6)	(4.9)	(11.0)	(7.2)	(3.5)
Others	1.0	6.7	0.5	6.2	(0.0)	(0.0)
OPN cash flow	18.1	20.0	27.5	50.9	96.5	139.4
Capex	(0.5)	(0.5)	(0.6)	(1.5)	(2.1)	(2.1)
Chg in investment	(0.1)	(26.0)	0.3	0.0	0.0	0.0
Others	(0.2)	3.5	(0.3)	(0.4)	0.0	0.0
Investing CF	(0.7)	(23.1)	(0.6)	(2.0)	(2.1)	(2.1)
Chg in debt	(0.4)	20.3	(2.8)	(8.1)	(12.0)	(12.0)
Equity raised (Repurchased)	(7.6)	(12.2)	(6.1)	(8.3)	(16.0)	(16.0)
Dividend paid	(7.6)	(9.8)	(11.1)	(12.5)	(19.5)	(39.1)
Others	(0.0)	(0.1)	(0.1)	(0.0)	0.0	0.0
Financing CF	(15.6)	(1.7)	(20.1)	(28.9)	(47.5)	(67.1)
FX & other adj	0.0	0.0	0.0	0.0	0.0	0.0
Chg in cash	1.8	(4.8)	6.8	20.0	46.8	70.2
Beginning cash	12.4	14.2	9.3	16.2	36.2	83.0
End cash	14.2	9.3	16.2	36.2	83.0	153.1
FCF	17.6	19.4	26.9	49.3	94.3	137.3

RATIO & PER SHARE DATA

FYE Oct ($)	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Gross margin (%)	75%	77%	79%	75%	69%	67%
EBITDA margin (%)	72%	79%	79%	75%	69%	66%
Op. Profit margin (%)	62%	60%	66%	67%	64%	63%
Net Profit margin (%)	51%	46%	53%	55%	53%	53%
ROE (%)	77%	52%	45%	60%	75%	69%
ROA (%)	25%	20%	20%	31%	46%	49%
Gross gearing (%)	165%	102%	82%	51%	26%	13%
Net gearing (%)	106%	88%	62%	19%	-22%	-48%
Asset turnover (x)	49%	37%	34%	46%	66%	80%
Leverage (x)	3.0	2.6	2.3	1.9	1.6	1.4
Payable days	120.8	119.1	135.4	106.9	94.6	97.1
Receivable days	32.1	26.9	33.0	39.9	40.6	41.1
Inventory days	76.4	55.3	53.9	55.2	52.9	54.2
EPS, diluted	3.3	1.2	4.8	9.6	19.4	27.4
BVPS	5.6	14.3	16.8	23.3	35.4	51.5
Net cash per share	-6.0	-12.6	-10.4	-4.3	7.7	24.5
SPS	8.4	10.9	13.2	21.7	40.9	56.6
FCFPS	4.1	4.1	5.5	10.1	19.3	28.1
Total returns (%)	1.0	1.3	1.0	1.2	2.0	3.2

Source: Broadcom, Aletheia/TAG

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